Reconsidering Fund Fees: Disclosure Salience and the Decision to Stay*

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Abstract

Mutual fund investors choose once but pay for years. Because ownership continues by default and expenses are deducted from assets rather than billed, investors may keep paying for a fund that is no longer worth its fee. I ask whether sending existing investors more salient expense information changes whether they remain invested. The SEC's Tailored Shareholder Report reform replaced lengthy reports with short, class-specific reports that put expenses first. Using staggered first-report timing across funds' fiscal calendars, I find that retail flows become more fee-sensitive. A one-standard-deviation higher expense ratio is associated with an additional annualized net-flow reduction of 3.4 percentage points. Institutional classes of the same funds, whose investors rely less on these reports, show no significant change. Redemptions of existing capital account for most of the early response. Fee sensitivity rises more where earlier reports made expenses harder to locate and following deterioration in relative performance. High-fee funds also show relative fee declines and move their portfolios further from benchmarks, consistent with competitive responses to greater fee scrutiny. These findings show that delivering an already-public cost in a salient form can weaken inertia without asking investors to choose again.

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1 Introduction

"Performance comes, performance goes. Fees never falter."
—Warren E. Buffett, *2017 shareholder letter*

Choosing a mutual fund is an active decision, while continuing to hold it need not be. Ownership continues by default, with expenses deducted from fund assets rather than billed. As with a subscription, an investor can keep paying without reconsidering whether the fund remains worth its cost. Such inertia can sustain payments for years even as the investor's needs and the available alternatives change. The recurring expenses subject to this decision are large in aggregate. Following French (2008), I estimate that investors in U.S. equity mutual funds alone paid roughly $21 billion more in operating expenses in 2022 than the same assets would have cost at a low-cost index-fund benchmark.[1]

This paper asks whether a disclosure reform can change investors' decisions to remain invested by making ongoing expenses more salient in the periodic reports they receive. I study the SEC's Tailored Shareholder Report (TSR) reform, which replaces lengthy annual and semiannual reports covering multiple funds and share classes with short, class-specific reports that emphasize expenses and performance. TSR makes expenses the first item of the report, showing the cost of a hypothetical $10,000 investment in the investor's own class in both dollars and percentage terms. This prominent display may prompt investors to revisit a comparison that continued ownership does not otherwise require.

Whether it does depends on why investors continue to hold. Expense ratios were already available on brokerage platforms and in prior reports. If investors already weigh the fee and remain because they judge the fund worth it, or because switching costs outweigh the gain, restating the fee should change little. If instead the report prompts them to revisit a neglected expense, some may withdraw capital or reduce further investment. These decisions

[1]The calculation applies raw annual expense ratios to monthly asset balances in CRSP. The resulting expense bill is $32.9 billion, compared with $11.5 billion at a 14-basis-point fee. This benchmark is the highest 2022 expense ratio among Vanguard Total Stock Market Index Fund's conventional share classes. Both active and index funds are included.

aggregate into fund flows. If reassessment shifts capital away from higher-fee classes relative to lower-fee classes, net flows should become more negatively related to expense ratios after TSR. I test this prediction conditional on performance and other fund characteristics, then examine whether the response operates through redemptions or purchases.

I collect annual and semiannual shareholder reports from Forms N-CSR and N-CSRS on SEC EDGAR and link them to monthly CRSP fund share-class data. I separately link fund-level purchases and redemptions from Form N-PORT filings on EDGAR to CRSP. The shareholder-report filings identify each class's first observed TSR. Although the compliance deadline (July 24, 2024) was common to all funds, their existing fiscal calendars spread first TSR issuance over several months. The analysis uses this timing variation to examine how the relation between flows and expenses changes around each class's first report.

I focus on retail share classes, where the report is most likely to reach an investor susceptible to inertia. Institutional classes serve as a benchmark. They received TSRs on the same schedule, and classes of the same fund hold the same portfolio. Their holders, however, are either institutions, which already monitor fees closely, or households investing through retirement plans. The latter group, including holders of Class R shares, typically receives disclosures prepared by plan sponsors rather than the fund's own report. Both cases predict a smaller response among institutional classes. Consistent with this prediction, flow–fee sensitivity increases significantly among retail classes after the first TSR, while the change among institutional classes is muted. The flat institutional response weighs against explanations that predict similar changes across both clienteles, such as a marketwide reassessment of expensive funds' expected performance.

The retail response is economically substantial. After a class's first TSR, monthly net flow falls by an additional 0.287 percentage points of assets for a one-standard-deviation (56.7-basis-point) higher expense ratio, roughly 3.4 percentage points per year. Estimated flow changes decline monotonically from the cheapest to the most expensive fee quintile. The increase in fee sensitivity is robust when comparing classes that hold the same portfolio

and when measuring flows relative to investment-category peers, addressing changes in the attractiveness of underlying investments and category-level demand. It also persists when the flow–fee relation is allowed to vary across quarters. The increase around first TSR issuance is significantly larger than around corresponding reporting months in 2021 and 2022, distinguishing it from ordinary reporting-cycle effects. Finally, the result holds in a stacked difference-in-differences design that uses only not-yet-treated classes as controls throughout each cohort's comparison window.

Net flows combine withdrawals from existing holdings with purchases by both current investors and new buyers. Separating these margins shows whether the response involves capital already invested or new investment. Because TSR is directed at existing investors, a redemption response is particularly relevant to the interpretation that the report prompts reassessment of an existing holding. I examine these margins using fund-level purchases and redemptions from Form N-PORT. These data pool a fund's retail and institutional classes, so I classify funds by each clientele's share of assets. For funds with more than 50% of assets in retail classes, the statistically significant redemption response accounts for about 79% of the early implied change in net-flow sensitivity during the first three months after the issue month. This share rises to 96% for funds with more than 90% retail assets, while the purchase response is smaller and statistically insignificant at both thresholds. In contrast, institutional-dominant funds have small and statistically insignificant estimates for both purchases and redemptions.

The concentration in early redemptions supports the interpretation that TSR prompts investors to reconsider the ongoing cost of holdings they already own and reduce the balances on which those expenses accrue. An accumulated holding can be reduced at once, so reassessment can generate a large initial redemption response. As the estimated redemption response weakens in later months, reduced purchases account for a larger share of the overall adjustment. This pattern is consistent with existing investors reducing contributions, although the purchase data also include investments by new buyers.

If the flow response reflects more salient expense disclosure, it should be larger where the earlier report made expenses harder to find. I measure whether the electronic pre-reform report lacked a direct hyperlink to expense information. I also count the fund and share-class entries on the expense page. The increase in fee sensitivity is larger for classes whose reports lacked a direct link and for those with more crowded expense pages, consistent with greater improvements in expense accessibility for these classes. Neither measure is significantly associated with pre-announcement expenses or performance ranks (Figure 7), which alleviates the concern that they mainly capture deliberate obfuscation by expensive or poorly performing funds.

Greater salience of expenses does not by itself determine whether investors regard a fund's expense ratio as justified. That assessment depends on the performance and services they expect in return, and recent relative performance is one input. I therefore examine whether the increase in fee sensitivity varies with recent deterioration or improvement in performance relative to investment-style peers. Relative to all other classes, the increase in fee sensitivity is larger for classes newly entering the bottom tercile of annual returns within their style and smaller for classes newly entering the top tercile. The pattern is consistent with investors judging ongoing expenses in light of recent relative performance.

High-fee funds can respond to more fee-sensitive investors by lowering fees or differentiating their portfolios. I study both margins from the SEC's October 2022 adoption announcement onward. In the first two full fiscal years after the announcement, the expense-ratio gap between the highest- and lowest-fee quintiles of retail classes narrows by 3.48 basis points relative to its pre-announcement trend, mainly through expenses other than 12b-1 distribution fees. The portfolio analysis divides the post-announcement period into transition and post-implementation stages, separated by mandatory issuance in July 2024. Among retail-dominant active domestic equity portfolios, the Active Share gap between the highest-fee quintile and the remaining quintiles widens by 1.45 percentage points after implementation relative to the pre-announcement period, with no significant increase during the transition.

These results are consistent with funds competing on both price and portfolio differentiation (Christoffersen and Musto, 2002; Cremers et al., 2016; Wahal and Wang, 2011).

The paper contributes to the literature on the effectiveness of consumer financial disclosure. Evidence on how existing bank customers respond to more salient information is mixed. Making overdraft fees salient reduces their incidence (Stango and Zinman, 2014), whereas salient information about better savings products generates little switching (Adams et al., 2021).[2] In experiments on initial mutual-fund allocations, a salient fee summary shifts allocations toward cheaper index funds (Choi et al., 2010), whereas a simplified prospectus does not detectably change portfolio choice (Beshears et al., 2011).[3] Inattention to fees may be more pronounced after purchase, when ownership continues by default and expenses accrue without a new investment decision. This paper provides field evidence that redesigning recurring shareholder reports strengthens fee sensitivity in the decision to retain existing holdings, even though the expense ratio was already public.

The closest study is by Kronlund et al. (2021), who find increased fee sensitivity of 401(k) flows after participants received side-by-side fee and performance disclosures for their plan's options. Those disclosures make cheaper alternatives immediately identifiable. TSR instead reports the class an investor already holds, without competing funds alongside it. Its staggered rollout with monthly flow data allows me to trace adjustment around each class's first TSR. The increase in fee sensitivity shows that disclosure about an incumbent holding can change investors' willingness to retain capital without supplying a comparison of alternatives. Linked report and transaction data let me examine how investors adjust existing holdings and how report design shapes that response, extending the evidence beyond side-by-side comparisons of investment alternatives.

[2]Consumer-credit studies find reduced payday borrowing after disclosure of accumulating fees (Bertrand and Morse, 2011), limited effects of credit-card interest-rate disclosures (Seira et al., 2017), and improved loan repayment without detectable changes in initial loan terms (Kulkarni et al., 2026).

[3]Related research documents weaker responses to ongoing expenses than to front-end loads (Barber et al., 2005) and persistent fee dispersion among funds with similar portfolios (Cooper et al., 2021). Work on fee visibility examines shrouding (Anagol and Kim, 2012) and complexity in higher-fee funds' disclosures (deHaan et al., 2021).

The paper also contributes to research on consumer inertia. Studies of health-plan choice and mortgage refinancing separate inattention from adjustment costs as sources of inaction and find a substantial role for inattention (Andersen et al., 2020; Heiss et al., 2021). Einav et al. (2025) find that subscription cancellations spike when a replaced payment card forces subscribers to renew actively, so attention shocks there arrive together with a forced decision. TSR leaves continued ownership as the default and requires no response. The findings show that recurring disclosure can weaken inertia without reducing adjustment costs or requiring an active renewal decision.

The remainder of the paper proceeds as follows. Section 2 describes the TSR reform and its rollout. Section 3 presents a simple framework, and Section 4 describes the data and variable construction. Section 5 reports the net-flow results and robustness checks. Section 6 separates purchases from redemptions and examines the roles of pre-TSR expense accessibility and performance history. Section 7 studies funds' fee and portfolio responses. Section 8 concludes.

2 Institutional Background

2.1 The Tailored Shareholder Report Reform

Mutual funds send investors annual and semiannual reports to help them monitor their investments. Before TSR, these reports often covered several funds and share classes. They combined performance discussions, expense information, portfolio holdings, and financial statements. SEC staff estimated that pre-reform annual reports averaged 134 pages and semiannual reports 116 pages (U.S. Securities and Exchange Commission, 2022).[4] Information about an investor's own holding could be difficult to locate.

The SEC adopted the Tailored Shareholder Report rule on October 26, 2022. The reform replaces lengthy reports with short, standardized reports specific to each share class. These reports are approximately three to four pages long and bring together expenses, performance, and portfolio information. Expenses are the first substantive item. They show the cost of a hypothetical $10,000 investment alongside the class's annualized expense ratio. Detailed financial statements and complete portfolio holdings remain available online and upon request.

Figure 1 illustrates the change for Class C shares of the Growth Fund of America. Before TSR, funds were required to include an expense example but had discretion over its placement and presentation (U.S. Securities and Exchange Commission, 2013). In the pre-TSR report, the expenses for Class C appear on page 30. The table covers 21 share classes, each shown under two return assumptions. Investors must identify Class C and read the relevant figures across several columns. The tailored report instead presents the dollar cost and annualized expense ratio for Class C in a single row on the first page. It excludes figures for other classes. The example shows how the reform reduces the effort of reaching the expense information and identifying the applicable figure.

[4]For the retail classes in the sample, the median pre-TSR report contains approximately 38,962 words (Table 2).

The reform also changes how the report reaches some investors. Under Rule 30e-3, funds could send a notice directing investors to an online report. TSR replaces this notice-only option with transmission of the tailored report itself.[5] The short, class-specific report provides a recurring occasion to review expenses already available elsewhere.

2.2 First Tailored Reports and Reporting Calendars

The amendments became effective on January 24, 2023, followed by an 18-month transition. Compliance became mandatory for reports transmitted on or after July 24, 2024. Although this deadline was common to all funds, their existing fiscal calendars spread the first tailored reports across months.[6]

Funds prepare annual and semiannual reports according to their fiscal year-ends. They must transmit them within 60 days after the reporting-period end. Form N-CSR must then be filed with the SEC within 10 days after transmission, allowing up to 70 days after the period end (U.S. Securities and Exchange Commission, 2022; 2025). I infer an issue month from each first TSR filing date to approximate when the report was transmitted to investors. Section 4.2 explains this assignment. Panel A of Figure 2 shows the distribution of these inferred issue months. First TSR filings before the compliance deadline were rare.

Panel B examines how closely the rollout follows funds' reporting calendars. I use each fund's latest fiscal year-end in CRSP before the compliance deadline. From that calendar, I identify the first reporting deadline on or after July 24, 2024, allowing 70 days after each fiscal half-year end. I apply the same issue-month convention to these projected deadlines and the observed filings. The calendar-implied month matches the month inferred from the first TSR filing for 99.4% of share classes. The close alignment indicates that first-report timing largely reflects pre-existing fiscal calendars rather than discretionary early adoption,

[5] Internet Appendix Table IA.3 reports supplementary comparisons based on documented early intent to use Rule 30e-3.

[6] For 99.5% of share classes in the sample, the fiscal-year-end month recorded at the first TSR matches that in the last report before the reform's October 2022 announcement. Consistent with this calendar stability, Panel A of Internet Appendix Figure IA.1 shows broadly similar seasonal patterns in filing activity before and after the reform.

alleviating concerns that funds chose earlier issuance in anticipation of investor responses.

3 A Simple Framework

The framework links investors' review of existing holdings to the sensitivity of fund flows to expenses. More frequent review can lead to greater withdrawals or larger reductions in planned contributions at higher-fee funds, making net flows more negatively related to expenses. It separates review from the search and allocation decisions that follow, allowing the two adjustment margins to face different costs.

Let a be the probability that an investor reviews the holding during a month. Review means processing available fee and performance information to reassess the investment, following the separation of attention and choice in Heiss et al. (2021) and Einav et al. (2025).[7] Earlier shareholder reports already provided opportunities for review. TSR can make those opportunities more effective through changes in presentation and delivery. In the framework, TSR raises the probability of review from a_0 to a_1. This comparative static holds valuations, alternative opportunities, cost distributions, balances, contributions, and the investor population fixed. The empirical response may also reflect changes in how investors understand or weigh expenses conditional on review.

Upon review, the investor assesses the fund's expected net value over the coming year, $u(e, r)$, where e is its annual expense ratio and r is recent net performance. This value reflects the investor's expectations about investment benefits and services, expressed as a net payoff per dollar invested. I assume $\eta \equiv -\partial u/\partial e > 0$, holding recent performance and other benefits fixed. Expenses reduce prospective net value even conditional on recent net returns when those returns are an imperfect signal of future performance.[8]

[7] Review can involve reconsidering a known expense or understanding it for the first time. Public availability of the expense ratio does not establish the investor's prior knowledge of it.

[8] For example, suppose recent net performance is $r = \theta - e + \varepsilon$, where persistent gross performance θ has prior mean μ. If the posterior mean of gross performance is $(1-\lambda)\mu + \lambda(r+e)$, then expected future net value, including services b_s, is $u = (1-\lambda)(\mu - e) + \lambda r + b_s$. The calculation adds e back to recent net returns when inferring gross performance and subtracts e when evaluating future net value. Thus, $\eta = 1 - \lambda > 0$

Consider an investor with a balance $B > 0$ in the fund and a planned contribution $b > 0$ during the month. Both amounts are evaluated over the same one-year horizon. Payoffs are linear in capital, with no within-period discounting.[9] Without review, the investor retains B and contributes b, possibly through an automatic instruction. Following review, each amount can remain in the current fund or move entirely to an alternative.

TSR reports the cost of the current holding but does not identify a replacement. An investor who needs to investigate alternatives compares the expected gain from doing so with the search cost. Let X be an alternative's expected net value per dollar over the same horizon. The investor's beliefs about X have distribution H. Search costs s and reveals X.

Moving the existing balance costs $K_R \geq 0$, while redirecting the current contribution costs $K_C \geq 0$. These implementation costs are separable. Realizing taxable gains can add to K_R (Ivković and Weisbenner, 2009), as can the effort of unwinding an investment or reluctance to reverse an earlier choice. Changing a contribution instruction can enter K_C. Their relative size is unrestricted. One investigation can inform both decisions, so its expected benefit is

$$V(u) = \mathbb{E}\Big[\big(B(X-u) - K_R\big)_+ + \big(b(X-u) - K_C\big)_+\Big], \tag{1}$$

where $(x)_+ = \max\{x, 0\}$. The two terms are the gains from moving the balance and redirecting the contribution, each net of its implementation cost. The investor searches if these expected gains cover the search cost, $s \leq V(u)$. After learning X, the search cost is sunk. The investor moves the balance if $B(X-u) > K_R$ and redirects the contribution if $b(X-u) > K_C$. Either decision, or both, can justify the initial investigation.

For investors requiring search, let search costs have distribution G, independently of X. Review is independent of search costs and locally independent of expenses. Define the implementation costs per dollar as $k_R = K_R/B$ and $k_C = K_C/b$. Conditional on search, the

when recent performance is an imperfect signal. This illustration holds the recurring expense fixed across the historical and prospective periods.

[9]Investors without a planned contribution face only the holding decision.

probability that an alternative justifies adjustment on margin j is $T_j(u) = 1 - H(u + k_j)$. The probability of adjustment conditional on review is therefore

$$p_j(e, r) = G(V(u))T_j(u), \qquad j \in \{R, C\}. \tag{2}$$

Adjustment requires both that searching be worthwhile and that the alternative found be sufficiently attractive. A higher expense ratio reduces the value of staying, increasing both the expected benefit of search and the probability that the alternative clears the adjustment hurdle.

Expected redemptions from this investor are aBp_R, while expected purchases are $b(1 - ap_C)$. Expected monthly net flow, relative to the initial balance, is

$$N^{\text{inc}} = \frac{b}{B} - a\left[p_R + \frac{b}{B}p_C\right].$$

With locally smooth distributions and densities g and h, holding balances and implementation costs fixed, the fee sensitivities satisfy

$$\frac{\partial p_j}{\partial e} = \eta\Big[g(V)(BT_R + bT_C)T_j + G(V)h(u + k_j)\Big] \geq 0, \tag{3a}$$

$$\Delta\left(\frac{\partial N^{\text{inc}}}{\partial e}\right) = -(a_1 - a_0)\left[\frac{\partial p_R}{\partial e} + \frac{b}{B}\frac{\partial p_C}{\partial e}\right] \leq 0. \tag{3b}$$

The first term in $\partial p_j/\partial e$ captures additional search, and the second captures additional adjustment after an alternative is found. More frequent review can make redemptions more positively related to expenses and purchases more negatively related to expenses, making incumbent net flows more fee-sensitive.[10] The response is stronger where expenses move investors across search or adjustment thresholds. More frequent review need not induce adjustment if investors still prefer the fund or the gains from moving do not cover the costs.

[10] If the report prompts a larger increase in review among investors in higher-fee funds, this reinforces the predicted increase in fee sensitivity.

Weighting incumbent responses by their initial balances gives their contribution to observed fund flows. Redemptions come from existing holdings, while purchases also include investments by new buyers. A redemption response connects increased fee sensitivity to decisions about capital already invested.

The relative importance of redemptions and reduced contributions remains an empirical question. The relevant implementation costs are costs per dollar, K_R/B and K_C/b. A larger existing balance can make redemption worthwhile even when its total implementation cost exceeds the cost of redirecting a contribution. Taxes and other liquidation costs can instead favor contribution redirection. When $B > b$, a redemption also moves more capital than a forgone contribution.

Clientele and prior expense accessibility can shape the increase in review, $a_1 - a_0$. This increase may be smaller when investors already monitor through other channels or when reports do not reach the household making allocation decisions. A prominent, class-specific report can have more scope to prompt review where the applicable expense was previously harder to locate. Holding responses conditional on review fixed, a larger $a_1 - a_0$ produces a larger increase in flow–fee sensitivity.

Performance history enters through the perceived value of staying, $u(e, r)$. When recent disappointment lowers this value, searching becomes more attractive and more alternatives clear the adjustment hurdles. Recent success can have the opposite effect. How these changes affect fee sensitivity depends on investors' positions relative to the search and adjustment thresholds, as captured by $\partial p_R/\partial e$ and $\partial p_C/\partial e$. The performance-history comparisons examine whether the response to expenses varies with this assessment.

4 Data and Variable Construction

4.1 Data Sources and Sample Construction

I link mutual fund shareholder reports from SEC EDGAR to share-class data from CRSP to examine how investor flows respond to the introduction of tailored reports. The sample covers July 2023–July 2025. The window begins one year before the July 2024 compliance deadline, covering a full seasonal cycle in fund flows while keeping the comparison close to the reform. It ends one year after the deadline, allowing time to observe flows following the rollout.[11] The share class is the unit of the main analysis because expenses and tailored reports are specific to the class held by the investor.

I collect annual and semiannual shareholder reports from Form N-CSR and N-CSRS filings. These filings provide the dates and contents of pre-TSR and tailored reports. From the earlier reports, I extract information on the organization of expense disclosures and the ease with which investors could locate the information relevant to their class.

A fund corresponds to an SEC series and may offer multiple share classes. I match the SEC registrant, fund-series, and share-class identifiers in filing headers to those provided by CRSP. For records without a share-class identifier, historical ticker information supplements the match. I associate the extracted reports with the matching filing and class. These matches identify each class's first observed tailored report and, when available, earlier reports that may cover several funds and share classes. CRSP provides monthly class assets and returns, annual expense ratios, investment categories, and fund and family characteristics.

I require each class to have both a linked pre-TSR report and a linked TSR. I exclude known money-market funds, ETF, ETN, and variable-annuity classes, focusing the analysis on conventional mutual funds. The all-class sample contains 19,741 share classes from 5,946 funds. The retail-focused sample contains 8,377 share classes from 4,594 funds.

[11] The retail Post × Fee coefficient changes little when the sample is restricted to January 2024–January 2025.

To distinguish purchases from redemptions, I use monthly transactions reported at the fund level on Form N-PORT. I link these records to CRSP using SEC series identifiers, supplemented by CRSP information on which share classes belong to each series. The assets of the matched share classes are then aggregated. Purchases exclude reinvested distributions, and redemptions are reported as positive amounts. Both are expressed as a percentage of prior-calendar-month fund assets. The N-PORT sample contains 100,597 fund-months from 4,930 funds over July 2023–July 2025. For Active Share, I combine CRSP mutual fund holdings with historical constituent weights for funds' disclosed benchmark indices. Tracking error combines net fund returns with dividend-inclusive own-index returns.

4.2 Variable Construction

Following the standard flow measure in Sirri and Tufano (1998), monthly net flow for share class i is asset growth net of investment returns:

$$Flow_{it} = 100 \times \frac{AUM_{it} - AUM_{i,t-1}(1 + r_{it})}{AUM_{i,t-1}}, \tag{4}$$

where AUM denotes class assets at month-end and r_{it} is the monthly net return in decimal form. Net flow is expressed as a percentage of assets at the previous month-end and combines purchases and redemptions.

$Fee_{i,t-1}$ is the annual CRSP expense ratio for the previous month, expressed in percentage points. A one-unit difference in Fee corresponds to a difference of 100 basis points in annual expenses.

I infer each share class's issue month from its first TSR filing. In the data, the day-of-month distribution of N-CSR filings is strongly bimodal, with concentrations near the beginning and end of the month. Motivated by this pattern and the permitted 10-day filing lag, I assign filings dated on days 1–14 to the preceding month. Filings dated on day 15 or later are assigned to the filing month. This convention groups late-month filings with early

filings in the following month. The assigned issue month approximates when the report is delivered to investors. The indicator $Post_{it}$ equals one beginning in the first month after this issue month. The issue month itself remains in the pre-period.

Because tailored reports also present fund performance, the regressions account for investors' responses to past returns when estimating changes in fee sensitivity. I interact the performance controls with $Post_{it}$, allowing the flow–performance relationship to change after the first tailored report.[12] These controls are the previous month's net return and the class's net-return rank over the preceding 12 calendar months relative to other classes in the same Lipper category.

Additional controls include log class assets, return volatility, log class age, log family assets, and log family age, along with their interactions with $Post_{it}$. Appendix A provides the variable definitions. All continuous variables are winsorized at the 1% level.

4.3 Measuring Pre-TSR Expense Accessibility

Pre-TSR reports varied substantially in how easily investors could reach and identify the expenses of their own holdings. I use filing breadth, the number of distinct SEC series covered by a filing, as a broad measure of how much fund-level material is bundled in one document. Two further measures capture access to the applicable expense more directly. I record whether the report provides a direct expense link and count the fund or class choices on the expense page. Figure 3 illustrates these two measures using reports from actively managed domestic equity funds for the same reporting period.

The direct-link measure records whether the electronic report provides a verified hyperlink to the expense information applicable to the investor's class. In Panel A, AMG Montrusco Bolton's contents page links directly to Class N expenses, whereas Congress Large Cap Growth lists the expense section without a hyperlink.

[12] Kaniel and Parham (2017) show that prominent newspaper rankings affect fund flows, while Evans and Sun (2021) show that a change in rating methodology alters investors' risk adjustment. These findings motivate allowing the flow–performance relation to change with disclosure.

The expense-page measure counts distinct fund/class entries across all expense tables on the page, using the publisher's page boundaries. In Panel B, the page containing Hartford Growth Opportunities Class C expenses presents 45 fund/class choices. Haverford Quality Growth Stock's page contains just one.

The two fee-specific measures capture successive obstacles to reviewing expenses. A direct link can lead to a crowded expense page, while a page with a single choice can still take effort to locate. Thus, TSR's prominent, class-specific expense presentation can produce a larger improvement where earlier reports lacked a direct link or displayed many choices. For each class, I construct all three measures from its latest eligible conventional N-CSR or N-CSRS filing. Section 6 examines how the increase in fee sensitivity varies with these obstacles and with filing breadth.

4.4 Summary Statistics

Table 1 reports summary statistics separately for retail and institutional share classes. The mean monthly net flow is -0.778% for retail classes and -0.292% for institutional classes. Net flows are negative in 72% of retail class-months and 60% of institutional class-months. Retail classes are more expensive. Their mean annual expense ratio is 120.0 basis points, compared with 69.2 basis points for institutional classes. The gap is consistent with evidence that funds sold to institutions and larger accounts tend to charge lower fees (Khorana et al., 2009). Institutional classes are larger as well: the median class-month has approximately \$75 million in lagged assets, compared with \$35 million for retail classes. Mean monthly net returns and return volatilities are closer across the two groups. Mean returns are 0.99% for retail classes and 1.06% for institutional classes, and mean monthly return volatility is 3.674% and 3.555%, respectively. Appendix A defines the variables used in the analysis.

Table 2 describes the features of the latest pre-TSR report for retail classes, including length and expense accessibility. The median pre-TSR report contains approximately 38,962 words. In contrast, the median word count per report in the first TSR filing is only 775. The

median class is linked to a filing covering 10 funds, and the median expense page contains eight fund or class choices. For 56.6% of classes, the filing has no verified direct link to the applicable expense information.

5 Investor Responses to Tailored Reports

This section examines whether net flows become more sensitive to expenses around each class's first tailored report. I compare the changes between retail and institutional classes, then examine the magnitude, timing, and robustness of the retail response.

5.1 Flow–Fee Sensitivity Change

I begin by estimating the change in flow–fee sensitivity for the full sample of share classes.

$$\begin{aligned} Flow_{it} = \beta\big(Post_{it} \times Fee_{i,t-1}\big) + \delta Post_{it} + \theta Fee_{i,t-1} \\ + X_{it}'\gamma + \big(Post_{it} \times X_{it}\big)'\eta + \widehat{\rho}_{pre}\big(Fee_{i,t-1} \times \tau_t\big) + \alpha_i + \lambda_{c(i,t),t} + \varepsilon_{it}. \end{aligned} \tag{5}$$

The coefficient of interest, β, measures the change in the flow–fee slope following the report, in monthly-flow percentage points per 100 basis points of annual expenses. If TSR increases investors' sensitivity to expenses, β should be negative, indicating a more negative change in flows for higher-fee classes relative to lower-fee classes. X_{it} contains the controls defined in Section 4.2. Their interactions with $Post_{it}$ allow the control slopes to change after the report.

To allow for the long-standing trend toward greater flow–fee sensitivity (Morningstar, 2023), I estimate the model with and without a pre-estimated linear fee trend. In the adjusted specification, τ_t is calendar time in years centered on June 2024. The fee trend $\widehat{\rho}_{pre}$ is estimated over July 2019–June 2024 and held fixed in the baseline regression. β then measures the change relative to this trend. The regression includes share-class fixed effects

α_i and Lipper-category-by-month fixed effects $\lambda_{c(i,t),t}$. Standard errors are clustered by fund family.

Table 3 shows that the relation between net flows and expenses becomes more negative around the introduction of tailored reports. Column (1) estimates Equation (5) without a fee trend. The coefficient on Post $\times$ Fee is -0.2908. Adjusting for the pre-estimated fee trend in column (2) mildly reduces the magnitude to -0.2156. Thus, for a class with annual expenses 100 basis points higher, the change in its monthly flow rate is 21.6 basis points more negative following the first TSR. Internet Appendix Table IA.1 reports the full set of coefficients.

5.2 Retail and Institutional Responses

The average response across all share classes may conceal important differences between retail and institutional classes. There are two reasons to expect a negligible or small increase in fee sensitivity among institutional classes. First, institutional investors are more likely to incorporate fees into their ongoing evaluation of funds. Consistent with this view, Evans and Fahlenbrach (2012) find that institutional investors are more sensitive to high fees than retail investors in a comparison of retail and institutional fund twins. For investors who already monitor expenses, more salient expense disclosure has less scope to prompt a reassessment of their holdings. Second, institutional holdings can separate the ultimate investor from the recipient of the fund's report. Pension plans and other intermediaries hold mutual fund shares on behalf of households. In participant-directed retirement plans such as 401(k) plans, households receive fee and performance disclosures from the plan administrator. They need not automatically receive each underlying fund's shareholder report.[13] In these arrangements, changing the fund's report may leave the materials routinely seen by the household unchanged. In terms of the framework in Section 3, both considerations

[13]Department of Labor rules require plan administrators to provide participants with comparative investment information, including fees and performance. Fund shareholder reports received by the plan may be furnished automatically or upon request. See 29 CFR 2550.404a–5(d)(1)–(4) and U.S. Department of Labor (2010).

suggest a smaller report-induced increase in review, $a_1 - a_0$, among institutional investors than among retail investors.

Columns (3) and (4) test whether the increase in fee sensitivity differs between retail and institutional classes. I estimate the model jointly for the two groups and interact Post × Fee with an indicator for retail classes. In column (4), the institutional Post × Fee coefficient is 0.0066 and insignificant, while the retail-minus-institutional interaction is −0.7156 and significant at the 1% level. The implied retail coefficient is −0.7090. The separate regressions in columns (5) and (6) show the same retail–institutional pattern. The institutional coefficient in column (5) is −0.2419 and statistically insignificant, while for retail classes, column (6) gives a coefficient of −0.5061, significant at the 1% level.

For retail classes, the estimate in column (6) translates into a 0.287-percentage-point more negative change in monthly net flow for a one-standard-deviation difference in annual expenses, or 56.7 basis points. Annualizing this differential gives 3.44 percentage points. For a share class with median sample assets of $35.4 million, the change in annual net flows is approximately $1.22 million lower than for a class with expenses 56.7 basis points lower.

Figure 4 shows when the retail and institutional responses diverge. It traces changes in the flow–fee slope around the first TSR issue month.[14] In Panel A, the retail coefficients show no evidence of a pre-report trend but fall immediately to −0.378 in the first post-report interval, months +1 and +2. The estimates then remain negative and statistically significant throughout the displayed post-report period, indicating a sustained increase in fee sensitivity. In contrast, Panel B shows no statistically significant pre-report pattern for institutional classes, and the post-report estimates fluctuate around zero without a comparable decline. The contrasting paths reinforce the retail–institutional difference estimated in Table 3.

The institutional comparison also helps assess whether the retail result reflects a broader shift in demand. A marketwide reassessment that made expensive active funds less attractive relative to low-cost index funds should shift both retail and institutional demand toward

[14] I group event months into two-month intervals to reduce noise. The issue month (month zero) is the reference period. The results are qualitatively similar with one-month intervals.

cheaper funds. The contrasting responses weigh against such a common reassessment as the explanation for the retail pattern.

The remaining share-class analyses focus on retail classes, where the report is more likely to reach the individual making the allocation decision.

5.3 Flow Responses across the Fee Distribution

I next examine whether the response extends across the fee distribution. Rather than imposing a linear relation between expenses and post-TSR flow changes, I estimate a separate Post change for each fee quintile. The quintiles are based on retail classes' average annual expenses from October 2021 through September 2022. The specification imposes no ordering across quintiles. Median annual expenses rise from 52 basis points in the lowest quintile to 198 basis points in the highest.

The point estimates in Figure 5 decline monotonically across the five fee quintiles. Estimated monthly flow changes are 0.458, 0.261, 0.185, −0.041, and −0.300 percentage points from the lowest to the highest quintile.[15] This ordered pattern supports the baseline finding of increased retail fee sensitivity without requiring the post-TSR response to be linear in expenses.

5.4 Alternative Empirical Comparisons

Table 4 shows that the increase in retail fee sensitivity is robust to within-portfolio and within-fund comparisons and to category-adjusted flows. It also holds with flexible calendar-time fee slopes, in comparisons with earlier reporting cycles, and in a stacked difference-in-differences design using only not-yet-treated classes as controls.

An expensive fund may lose investors because its investments have become less attractive, even if investors have not become more sensitive to its expense. To address this concern,

[15]The estimates are conditional Post changes evaluated at annual-return rank 0.5 and the model's mean controls. A positive change indicates an improvement relative to the model's pre-report benchmark. It does not require positive net flows in levels.

column (1) replaces category-by-month fixed effects with portfolio-by-month fixed effects. The comparison is now between retail share classes holding the same portfolio in the same month, absorbing changes in the underlying investments and their gross performance. The Post $\times$ Fee coefficient is -0.9168. Because a CRSP portfolio can encompass more than one fund, column (2) uses the more granular fund-by-month fixed effects, comparing classes within the same reporting fund in the same month. The estimate is similar at -0.8967. These results show that the negative shift also appears among classes sharing the same underlying investments.[16]

Column (3) examines flows relative to competing funds in the same investment category. I subtract the contemporaneous, lagged-asset-weighted flow of the other classes in the category from each class's flow and include calendar-month fixed effects. This specification measures a class's ability to attract or retain assets relative to an asset-weighted peer benchmark, rather than absorbing category-by-month effects in the unweighted class-level regression. The Post $\times$ Fee coefficient is -0.6294. Higher-fee classes also experience a more negative flow change relative to their peers.

Column (4) addresses the assumption that the pre-reform fee trend provides a useful benchmark during the rollout. Investor preferences for low-cost funds may evolve with market conditions in ways that a linear extrapolation does not capture. The specification replaces the pre-estimated trend with fee-by-calendar-quarter interactions, allowing the marketwide flow–fee relation to change freely across quarters. The Post $\times$ Fee coefficient remains negative at -0.3043. Thus, the result persists when the fee gradient can follow a nonlinear calendar pattern within the estimation period. Since these quarterly slopes can also absorb a common response to TSR, this coefficient captures a narrower part of the change than the baseline estimate.

Column (5) addresses the concern that receiving any shareholder report, even in the previous format, could prompt investors to reconsider expenses and generate the observed

[16]These within-group comparisons use a smaller sample with multiple retail classes, so their larger coefficients are not directly comparable to the baseline estimate.

increase in flow–fee sensitivity. The test compares the change around the first TSR with the average change around the corresponding reporting months in 2021 and 2022. Given the stability of funds' reporting calendars, I approximate these earlier report months by shifting the first-TSR issue calendar back by two and three years.[17] I estimate the three episodes jointly, allowing each its own fixed effects, control slopes, and preceding flow–fee trend. The actual Post–fee change is 0.6028 percentage points more negative than the average historical change. This result alleviates the concern that the usual arrival of shareholder reports alone accounts for the response observed after TSR.

Finally, column (6) examines whether the increase in fee sensitivity is robust to excluding already-treated classes from the comparison group. With staggered treatment, conventional fixed-effects regressions can use earlier-treated units as controls for later-treated units. These "forbidden comparisons" can distort estimates when treatment effects evolve over time (Baker et al., 2022). I use a stacked difference-in-differences design that preserves the flow–fee specification while restricting the comparison group to later issuers.[18]

Each stack covers months -12 through $+3$ relative to the first TSR issuance month. Control classes must issue their first TSR after the end of the stack's post window. The compressed rollout leaves later cohorts with few eligible controls and little overlap between treated and control fee distributions. For this reason, I retain the July and August 2024 issue cohorts for this three-month comparison. I pool the two stacks with class-by-stack and category-by-month-by-stack fixed effects, so already-issued classes never serve as controls within a stack.

[17] I shift the calendar back by at least two years to avoid overlap between the historical and actual estimation windows. For this comparison, the actual window is July 2023–June 2025, and the historical windows are July 2021–June 2023 and July 2020–June 2022. Using two historical episodes reduces sensitivity to fluctuations specific to a single year.

[18] The standard estimators of Callaway and Sant'Anna (2021) and Sun and Abraham (2021) cannot directly estimate the change in the flow–fee slope. Applied to net flows, they estimate an average treatment effect. Recovering the slope change requires estimating how that response varies with fees.

The stacked estimate remains negative at −0.3358 and significant. Its magnitude is smaller than the retail baseline estimate of −0.5061 in Table 3, column (6). It is close, however, to the −0.378 estimate for the first two post-report months in Figure 4.

6 Reconsidering the Cost of Existing Holdings

Section 5 documents an increase in retail fee sensitivity. This section examines whether that response is consistent with investors reconsidering the ongoing cost of existing holdings. I separate redemptions from purchases to test whether the increase in fee sensitivity comes from investors withdrawing capital already invested in the fund. The accessibility comparisons test whether fee sensitivity rises more where earlier reports made expenses harder to find. The performance-history comparisons examine whether recent performance shapes investors' willingness to keep paying.

6.1 Redemptions of Existing Holdings

TSR is sent to current investors, so redemptions provide the most direct test of whether the report changes their willingness to retain existing holdings. Purchases combine money from new buyers with additional investment by existing investors. For existing investors, reducing future purchases offers a way to respond without selling their current holdings. The adjustment costs in Section 3 can favor this response, particularly when selling would trigger capital gains taxes (Ivković and Weisbenner, 2009).

I use monthly Form N-PORT transactions to examine redemptions and purchases at the fund level. These data aggregate transactions across a fund's retail and institutional share classes and provide complementary evidence on the two adjustment margins. Given the stronger response among retail classes in Section 5.2, I distinguish retail- and institutional-dominant funds using a share of June 2024 assets above 50%, with a 90% threshold identifying more concentrated clienteles. Figure 6 shows changes in each margin's relation with fees over

the first three months after the issue month and from month four onward. Redemptions are reported as positive amounts, and purchases exclude reinvested distributions. Both are scaled by prior-month fund assets. A positive redemption coefficient or a negative purchase coefficient implies a more negative net-flow response for higher-fee funds. Internet Appendix Table IA.2 reports the full estimates.

In Figure 6, the blue bars show the redemption coefficients and the gray hatched bars show the purchase coefficients. In Panel (a), the blue bars extend above zero and the gray bars below zero, so their combined length gives the magnitude of the negative implied net-flow effect. For funds with more than 50% of assets in retail classes, the redemption coefficient is 0.796 and the purchase coefficient is only −0.211. At the 90% retail threshold, the redemption coefficient is larger at 1.343, while the purchase coefficient is closer to zero at −0.061. Neither purchase coefficient is statistically significant. These point estimates place most of the change in the net-flow–fee relation on redemptions, with a larger redemption share at the higher retail threshold.

Panel (b) provides a contrasting picture for institutional-dominant funds. The redemption coefficients are −0.167 and −0.007 at the 50% and 90% thresholds, and the purchase coefficients are −0.142 and 0.028. None is statistically significant. At the 50% threshold, the small negative coefficients on the two margins largely offset. At the 90% threshold, both estimates are close to zero. Both comparisons imply little net-flow change. The figure shows a pronounced retail redemption response alongside small and statistically insignificant institutional estimates on both margins.

Panels (c) and (d) show the responses from month four onward. Among retail-dominant funds, the redemption point estimates become smaller while the purchase point estimates become more negative. Reduced purchases account for a larger fraction of the implied net-flow response, rising from 21% to 46% at the 50% threshold and from 4% to 31% at the 90% threshold. Redemptions remain the larger component, but the later response is more evenly distributed between the two margins. For institutional-dominant funds, both margins

remain small and statistically insignificant, with little implied net-flow response.

An accumulated holding can be withdrawn at once, whereas redirecting contributions reduces purchases as payments come due. This difference offers one interpretation of the larger early redemption response and the growing contribution of reduced purchases later.

The results in this subsection show that in retail-dominant funds, redemptions account for most of the early response, consistent with TSR prompting investors to act on the ongoing cost of holdings that continue by default.

6.2 Prior Report Design and Expense Salience

If greater expense salience contributes to the flow response, the increase in fee sensitivity should be larger where TSR changes the presentation most. The reform introduces a common, class-specific expense display, but the improvement depends on the report it replaces. I use the pre-TSR expense accessibility measures constructed in Section 4.3 to capture this variation.

One concern is that expensive or poorly performing funds make expenses harder to find, so the accessibility measures also capture differences in fund characteristics. Figure 7 plots the correlations of the three measures of fee accessibility with pre-announcement expenses and performance. Panel A shows annual expenses, and Panel B shows the prior 12-month category-relative return percentile. None of the six associations is statistically significant. These comparisons alleviate the concern that difficult reports simply identify low-quality funds.

Table 5 then tests whether fee sensitivity increases more where earlier reports made expenses harder to find. The Post $\times$ Fee $\times$ Low accessibility coefficient measures the additional flow–fee slope change for the less-accessible group relative to the remaining classes.

Fee sensitivity increases more in the less-accessible group under all three measures. In column (1), the additional slope change is -0.476 for classes in the top tercile of filing breadth. It is -0.578 for classes without a verified applicable fee link in column (2) and

−0.705 for the top tercile of expense-page choices in column (3).

Fee sensitivity rises more where TSR should increase expense salience most. This pattern supports the salience channel of TSR.

6.3 Recent Performance Changes and Fee Sensitivity

Recent performance can shape investors' willingness to keep paying a fund's ongoing expenses. A deterioration in relative performance can strengthen fee sensitivity, while an improvement can moderate it.

The baseline regressions already control for lagged monthly returns and annual net-return ranks and their Post interactions. Here, I test whether recent deterioration or improvement in relative performance also shapes the increase in fee sensitivity. Table 6 groups retail classes using annual net-return ranks within Lipper categories over July 2022–June 2023 and July 2023–June 2024. Columns (1)–(2) examine classes that enter the bottom tercile in the more recent year after being outside it in the earlier year. Columns (3)–(4) examine classes that enter the top tercile after being outside it in the earlier year. Each group is compared with all other retail classes. Within each pair, the first column omits the retail pretrend adjustment and the second includes it. The interaction with Entry measures the additional change in the flow–fee slope for the classes entering the indicated tercile.

In column (2), the Entry interaction is −0.354 for classes entering the bottom tercile. In column (4), it is 0.323 for classes entering the top tercile. Fee sensitivity increases more following a deterioration in relative performance and less following an improvement. The pattern is consistent with investors judging ongoing expenses against the performance they have recently received.

7 Funds' Response to Competition

As existing investors become more sensitive to expenses, expensive funds face greater pressure to justify the cost of staying. They can lower that cost or differentiate their portfolios from cheaper alternatives. Prior research documents both margins of competition. Wahal and Wang (2011) find that entry reduces incumbents' management fees, while Cremers et al. (2016) link stronger index-fund competition to lower fees and more active management. Sun (2021) finds that incumbents increase active management following low-cost index-fund entry, with fee responses differing across distribution channels. The analysis examines these two margins around TSR. Because managers can anticipate greater scrutiny before investors receive their first tailored report, I date these comparisons from the SEC's adoption announcement on October 26, 2022.

7.1 Fee Response

Table 7 examines whether expensive retail classes reduce their expense ratios relative to cheaper classes. Unlike the monthly flow analysis, this comparison uses an annual share-class panel, with one observation for each annual fiscal period. I assign classes to fee quintiles using mean expense ratios observed from October 2021 through September 2022, separately for active and passive funds within each investment style, and hold those assignments fixed. The reported estimates compare the highest- and lowest-fee quintiles. The sample covers annual fiscal-period endpoints in 2020–2025. The post-announcement period pools the first two full fiscal years starting after October 26, 2022. Earlier fiscal years, including the one containing the announcement, form the reference period.

The regressions include share-class and style-by-fiscal-endpoint-year fixed effects, together with annual counterparts to the baseline fund characteristics. These controls are measured before each annual fee period begins and enter both separately and in interactions with the post-announcement indicator. I also account for differences in the quintiles' earlier fee

trajectories by subtracting relative linear trends estimated over the five years before the announcement.

The expense gap is 3.48 basis points below its projected path in the post-announcement period. This amounts to about 2.9% of the average pre-announcement gap of 119.2 basis points. The post-announcement adjustment is concentrated in nonmarketing expenses, with an estimated change of -4.19 basis points. In contrast, the 12b-1 fee estimate is positive and insignificant. These results indicate that high-fee retail classes narrow their fee premium over cheaper classes more sharply than their pre-announcement trends would predict.

7.2 Product Response

A portfolio that closely tracks a low-cost index offers investors less reason to continue paying a high fee. For this analysis, I restrict the sample to active domestic equity funds. Their managers have discretion to deviate from a benchmark, and detailed U.S. stock-holdings and benchmark data allow those deviations to be measured. I examine whether expensive funds respond to increased price scrutiny by moving further from their benchmarks. Active Share captures differences in company weights within normalized common-stock holdings. Tracking error captures variation in whole-fund net returns relative to the fund's own index.

The unit of observation is the investment portfolio, which can be shared by multiple funds and their share classes. Portfolios can include both retail and institutional classes. I classify a portfolio as retail-dominant or institutional-dominant when the corresponding classes account for more than 50% of its September 2022 assets. I require known class designations for at least 80% of portfolio assets.

Each portfolio's pre-announcement expense ratio is an average across its share classes, weighted by September 2022 assets. Within each investment style, I rank retail-dominant and institutional-dominant portfolios separately using this measure. Table 8 compares changes in Active Share and tracking error between the highest-fee quintile and the other four quintiles.

Quarterly holdings and monthly returns allow a finer timing comparison than annual fee

data. Managers may reposition in anticipation of greater fee scrutiny, or wait until investors receive the reports and a flow response becomes apparent. I divide the post-announcement period into a transition stage before mandatory issuance (2022Q4–2024Q2) and a post-implementation stage (2024Q3–2025Q3). The pre-announcement period (2021Q3–2022Q3) serves as the reference.

All regressions include portfolio and style-by-time fixed effects. Following Sun (2021), the controls include log fund size, log fund age, and log family size. Fund and family assets are lagged one quarter for quarterly Active Share and one month for monthly tracking error. Portfolio age is measured at the observation date. Appendix A provides the variable definitions.

Among retail-dominant portfolios, the Active Share gap between the highest-fee quintile and the remaining quintiles rises by 1.446 percentage points in the post-implementation period. The transition estimate is only 0.305 percentage points and is statistically insignificant. The Active Share gap widens significantly only after mandatory TSR issuance begins. The tracking-error gap widens earlier, by 0.564 annualized percentage points during the transition and 0.750 after implementation.[19]

If managers adjust portfolios in response to greater fee sensitivity among retail investors, the adjustment should be stronger among retail-dominant portfolios. The point estimates follow this pattern. Among institutional-dominant portfolios, the post-implementation Active Share estimate is 0.288 percentage points and statistically insignificant, while the tracking-error gap widens significantly by 0.526 annualized percentage points. Both estimates are smaller than their retail counterparts, but neither retail–institutional difference is statistically significant. The comparison thus provides suggestive evidence that portfolio adjustment is stronger among portfolios more exposed to retail demand.

[19]For scale, the 1.446-percentage-point Active Share estimate equals 1.8% of the pre-announcement median level of 80.3% among retail-dominant portfolios. The annualized tracking-error estimates of 0.564 and 0.750 percentage points equal 11.3% and 15.0% of its pre-announcement median level of 4.99%, respectively.

Together, the results are consistent with managers responding to greater expense scrutiny through both fees and portfolio differentiation. Lower relative fees reduce the cost of staying, while greater differentiation changes what investors receive for that cost.

8 Conclusion

Disclosure can matter for the capital a fund retains as well as the capital it attracts. TSR puts mutual fund operating expenses first in short, class-specific reports to existing investors, showing costs in dollars and percentage terms. After the reform, retail net flows become more sensitive to expenses, while institutional classes show no significant change. The retail pattern spans the fee distribution and holds in within-portfolio comparisons, with flexible calendar-time fee slopes, and with not-yet-treated controls.

Redemptions account for most of the early response among retail-dominant funds. Reduced purchases account for a larger share of the later response. Fee sensitivity rises more where earlier reports made the applicable expense harder to reach or identify, consistent with a larger increase in expense salience. It also rises more following deterioration in relative performance and less following improvement. These patterns are consistent with investors revisiting ongoing expenses and judging them against recent performance.

The investor response is accompanied by adjustments on the fund side. Expensive retail classes reduce their fee premium beyond what their pre-announcement trends predict, mainly through nonmarketing expenses. Among retail-dominant active domestic equity portfolios, the expensive group also increases Active Share relative to other portfolios after mandatory implementation.

As in subscription markets, continued payment need not reflect a fresh assessment of value. The findings indicate that recurring communication can weaken inertia even when the price is already public, continued ownership remains the default, and no response is required. Whether the resulting reallocations benefit investors depends on their subsequent holdings, fees, risk, and net performance, including the costs of switching.

References

Adams, P., Hunt, S., Palmer, C., and Zaliauskas, R. (2021). Testing the effectiveness of consumer financial disclosure: Experimental evidence from savings accounts. *Journal of Financial Economics*, 141(1):122–147.

Anagol, S. and Kim, H. H. (2012). The impact of shrouded fees: Evidence from a natural experiment in the Indian mutual funds market. *American Economic Review*, 102(1):576–593.

Andersen, S., Campbell, J. Y., Nielsen, K. M., and Ramadorai, T. (2020). Sources of inaction in household finance: Evidence from the Danish mortgage market. *American Economic Review*, 110(10):3184–3230.

Baker, A. C., Larcker, D. F., and Wang, C. C. Y. (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics*, 144(2):370–395.

Barber, B. M., Odean, T., and Zheng, L. (2005). Out of Sight, Out of Mind: The Effects of Expenses on Mutual Fund Flows. *The Journal of Business*, 78(6):2095–2120.

Bertrand, M. and Morse, A. (2011). Information disclosure, cognitive biases, and payday borrowing. *The Journal of Finance*, 66(6):1865–1893.

Beshears, J., Choi, J. J., Laibson, D., and Madrian, B. C. (2011). How does simplified disclosure affect individuals' mutual fund choices? In Wise, D. A., editor, *Explorations in the Economics of Aging*, pages 75–96. University of Chicago Press.

Buffett, W. E. (2018). 2017 Letter to Shareholders. Berkshire Hathaway Inc. February 24.

Callaway, B. and Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.

Choi, J. J., Laibson, D., and Madrian, B. C. (2010). Why does the law of one price fail? An experiment on index mutual funds. *The Review of Financial Studies*, 23(4):1405–1432.

Christoffersen, S. E. K. and Musto, D. K. (2002). Demand curves and the pricing of money management. *The Review of Financial Studies*, 15(5):1499–1524.

Cooper, M. J., Halling, M., and Yang, W. (2021). The persistence of fee dispersion among mutual funds. *Review of Finance*, 25(2):365–402.

Cremers, M., Ferreira, M. A., Matos, P., and Starks, L. (2016). Indexing and active fund management: International evidence. *Journal of Financial Economics*, 120(3):539–560.

deHaan, E., Song, Y., Xie, C., and Zhu, C. (2021). Obfuscation in mutual funds. *Journal of Accounting and Economics*, 72(2):101429.

Einav, L., Klopack, B., and Mahoney, N. (2025). Selling subscriptions. *American Economic Review*, 115(5):1650–1671.

Evans, R. B. and Fahlenbrach, R. (2012). Institutional investors and mutual fund governance: Evidence from retail–institutional fund twins. *The Review of Financial Studies*, 25(12):3530–3571.

Evans, R. B. and Sun, Y. (2021). Models or stars: The role of asset pricing models and heuristics in investor risk adjustment. *The Review of Financial Studies*, 34(1):67–107.

French, K. R. (2008). Presidential address: The cost of active investing. *The Journal of Finance*, 63(4):1537–1573.

Heiss, F., McFadden, D., Winter, J., Wuppermann, A., and Zhou, B. (2021). Inattention and switching costs as sources of inertia in Medicare Part D. *American Economic Review*, 111(9):2737–2781.

Ivković, Z. and Weisbenner, S. (2009). Individual investor mutual fund flows. *Journal of Financial Economics*, 92(2):223–237.

Kaniel, R. and Parham, R. (2017). WSJ Category Kings: The impact of media attention on consumer and mutual fund investment decisions. *Journal of Financial Economics*, 123(2):337–356.

Khorana, A., Servaes, H., and Tufano, P. (2009). Mutual fund fees around the world. *The Review of Financial Studies*, 22(3):1279–1310.

Kronlund, M., Pool, V. K., Sialm, C., and Stefanescu, I. (2021). Out of sight no more? the effect of fee disclosures on 401(k) investment allocations. *Journal of Financial Economics*, 141(2):644–668.

Kulkarni, S., Truffa, S., and Iberti, G. (2026). Removing the fine print: Disclosure, standardized products, and consumer outcomes. *Journal of Financial Economics*, 185:104353.

Morningstar (2023). Morningstar finds investors saved nearly $9.8 billion in fund fees in 2022. Press release on the 2022 U.S. Fund Fee Study, August 8.

Seira, E., Elizondo, A., and Laguna-Müggenburg, E. (2017). Are information disclosures effective? evidence from the credit card market. *American Economic Journal: Economic Policy*, 9(1):277–307.

Sirri, E. R. and Tufano, P. (1998). Costly Search and Mutual Fund Flows. *The Journal of Finance*, 53(5):1589–1622.

Stango, V. and Zinman, J. (2014). Limited and varying consumer attention: Evidence from shocks to the salience of bank overdraft fees. *The Review of Financial Studies*, 27(4):990–1030.

Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.

Sun, Y. (2021). Index fund entry and financial product market competition. *Management Science*, 67(1):500–523.

U.S. Department of Labor (2010). Fiduciary requirements for disclosure in participant-directed individual account plans. Federal Register, 75(202), 64910–64946. Final rule, October 20, 29 CFR 2550.404a–5.

U.S. Securities and Exchange Commission (2013). How to read a mutual fund shareholder report. Investor Bulletin, SEC Publication No. 138.

U.S. Securities and Exchange Commission (2022). Tailored shareholder reports for mutual funds and exchange-traded funds; fee information in investment company advertisements. Investment Company Act Release No. 34731. Final rule, SEC release.

U.S. Securities and Exchange Commission (2025). Tailored shareholder reports frequently asked questions. Question 10, SEC guidance.

Wahal, S. and Wang, A. Y. (2011). Competition among mutual funds. *Journal of Financial Economics*, 99(1):40–59.

Figure 1: Expense disclosure before and after TSR

(a) Pre-TSR report, page 30

Expense example (continued)

	Beginning account value 9/1/2023	Ending account value 2/29/2024	Expenses paid during period*	Annualized expense ratio
Class A – actual return	$ 1,000.00	$ 1,189.56	$ 3.38	.62%
Class A – assumed 5% return	1,000.00	1,021.78	3.12	.62
Class C – actual return	1,000.00	1,185.29	7.50	1.38
Class C – assumed 5% return	1,000.00	1,018.00	6.92	1.38
Class T – actual return	1,000.00	1,191.28	1.85	.34
Class T – assumed 5% return	1,000.00	1,023.17	1.71	.34
Class F-1 – actual return	1,000.00	1,189.32	3.65	.67
Class F-1 – assumed 5% return	1,000.00	1,021.53	3.37	.67
Class F-2 – actual return	1,000.00	1,190.88	2.18	.40
Class F-2 – assumed 5% return	1,000.00	1,022.87	2.01	.40
Class F-3 – actual return	1,000.00	1,191.42	1.63	.30
Class F-3 – assumed 5% return	1,000.00	1,023.37	1.51	.30
Class 529-A – actual return	1,000.00	1,189.40	3.59	.66
Class 529-A – assumed 5% return	1,000.00	1,021.58	3.32	.66
Class 529-C – actual return	1,000.00	1,184.98	7.66	1.41
Class 529-C – assumed 5% return	1,000.00	1,017.85	7.07	1.41
Class 529-E – actual return	1,000.00	1,188.10	4.84	.89
Class 529-E – assumed 5% return	1,000.00	1,020.44	4.47	.89
Class 529-T – actual return	1,000.00	1,190.76	2.29	.42
Class 529-T – assumed 5% return	1,000.00	1,022.77	2.11	.42
Class 529-F-1 – actual return	1,000.00	1,190.57	2.45	.45
Class 529-F-1 – assumed 5% return	1,000.00	1,022.63	2.26	.45
Class 529-F-2 – actual return	1,000.00	1,190.79	2.18	.40
Class 529-F-2 – assumed 5% return	1,000.00	1,022.87	2.01	.40
Class 529-F-3 – actual return	1,000.00	1,191.06	1.91	.35
Class 529-F-3 – assumed 5% return	1,000.00	1,023.12	1.76	.35
Class R-1 – actual return	1,000.00	1,185.05	7.50	1.38
Class R-1 – assumed 5% return	1,000.00	1,018.00	6.92	1.38
Class R-2 – actual return	1,000.00	1,184.99	7.55	1.39
Class R-2 – assumed 5% return	1,000.00	1,017.95	6.97	1.39
Class R-2E – actual return	1,000.00	1,186.81	5.98	1.10
Class R-2E – assumed 5% return	1,000.00	1,019.39	5.52	1.10
Class R-3 – actual return	1,000.00	1,187.73	5.11	.94
Class R-3 – assumed 5% return	1,000.00	1,020.19	4.72	.94
Class R-4 – actual return	1,000.00	1,189.35	3.48	.64
Class R-4 – assumed 5% return	1,000.00	1,021.68	3.22	.64
Class R-5E – actual return	1,000.00	1,190.61	2.40	.44
Class R-5E – assumed 5% return	1,000.00	1,022.68	2.21	.44
Class R-5 – actual return	1,000.00	1,191.35	1.85	.34
Class R-5 – assumed 5% return	1,000.00	1,023.17	1.71	.34
Class R-6 – actual return	1,000.00	1,191.53	1.63	.30
Class R-6 – assumed 5% return	1,000.00	1,023.37	1.51	.30

(b) Tailored report, first page

ANNUAL SHAREHOLDER REPORT

The Growth Fund of America®

Class C | GFACX for the year ended August 31, 2024

This annual shareholder report contains important information about The Growth Fund of America (the "fund") for the period from September 1, 2023 to August 31, 2024. You can find additional information about the fund at capitalgroup.com/mutual-fund-literature-C. You can also request this information by contacting us at (800) 421-4225.

What were the fund costs for the last year? (based on a hypothetical $10,000 investment)

Share class	Costs of a $10,000 investment	Costs paid as a percentage of a $10,000 investment
Class C	$156	1.36%

Management's discussion of fund performance

The fund's Class C shares gained 29.09% for the year ended August 31, 2024. That result compares with a 27.14% gain for the S&P 500 Index. For information on returns for additional periods, including the fund lifetime, please refer to capitalgroup.com/mutual-fund-returns-C.

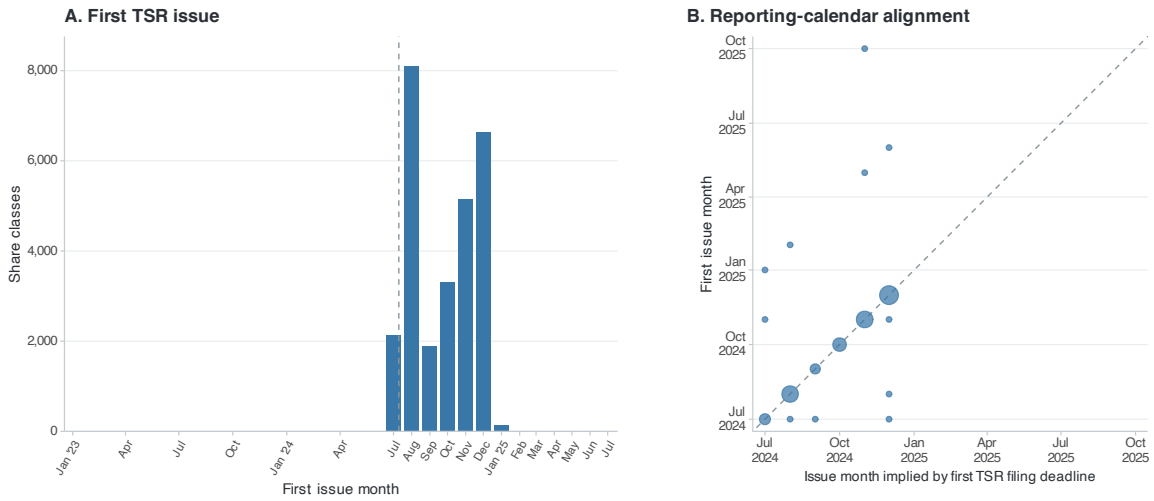
What factors influenced results

The U.S. economy continued to see growth during the fund's fiscal year, despite high interest rates. However, lower job creation numbers over the summer were seen as a sign of a possible interest rate cut by the U.S. Federal Reserve, thus giving investors hope that recession could be avoided. Overseas markets remained mixed during the period, though equity markets in the U.S. and elsewhere continued to see strong growth, albeit with bouts of volatility.

Holdings in most equity sectors produced positive returns for the fund, with shares of communication services, information technology, financials and utility companies particularly additive to results. While the bulk of the portfolio remains invested in U.S. equities, holdings of companies based in Taiwan, the Netherlands and Canada added to returns.

Notes: This figure compares expense disclosures for the Growth Fund of America, Class C (GFACX). Panel (a) shows the last pre-TSR report for the period ended February 29, 2024, filed April 30, 2024. Panel (b) shows the first TSR for the period ended August 31, 2024, filed October 31, 2024.

Figure 2: Timing of first tailored shareholder reports



Notes: This figure shows the distribution of first tailored-report issue months and their alignment with projected filing deadlines. Panel A shows first tailored-report issue months for linked share classes from January 2023 through July 2025. The dashed line marks the July 24, 2024 compliance deadline. Panel B compares the first issue month with the issue month implied by the projected deadline for filing the first tailored report. Circle area reflects the number of classes, with a minimum size for visibility; the diagonal marks equality. The projected deadline is the first semiannual reporting-period end plus 70 days on or after the compliance date, using the latest pre-deadline fiscal year-end. Filing and projected deadline dates on days 1–14 map to the preceding month, and later dates to the same month. The few large discrepancies include a delayed filing and cases in which an earlier class-specific tailored report was not established in the linked data.

Figure 3: Variation in expense accessibility across pre-TSR reports

Panel A. Direct link to the expense information

Direct link present: AMG Montrusco Bolton Large Cap Growth (MCGFX)

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AMG Funds
Semi-Annual Report — April 30, 2022 *(unaudited)*

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Notes: This figure illustrates variation in the accessibility of expense information in pre-TSR reports. Panel A shows the contents pages of semiannual reports for the period ended April 30, 2022. Outlines mark the expense entries. AMG's entry is a hyperlink to the applicable expense section; Congress's entry is a printed reference to page 37 without a hyperlink.

Figure 3: Variation in expense accessibility across pre-TSR reports (continued)

Panel B. Choices on the expense page

High choice count: 45 — Hartford Growth Opportunities, Class C (HGOCX)

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Hartford Domestic Equity Funds

Expense Examples (Unaudited) – (continued)

	Actual Return			Hypothetical (5% return before expenses)			
	Beginning Account Value November 1, 2021	Ending Account Value April 30, 2022	Expenses paid during the period November 1, 2021 through April 30, 2022	Beginning Account Value November 1, 2021	Ending Account Value April 30, 2022	Expenses paid during the period November 1, 2021 through April 30, 2022	Annualized expense ratio
The Hartford Dividend and Growth Fund							
Class A	$ 1,000.00	$ 979.20	$ 4.66	$ 1,000.00	$ 1,020.08	$ 4.76	0.95%
Class C	$ 1,000.00	$ 975.60	$ 8.48	$ 1,000.00	$ 1,016.22	$ 8.65	1.73%
Class I	$ 1,000.00	$ 980.50	$ 3.44	$ 1,000.00	$ 1,021.32	$ 3.51	0.70%
Class R3	$ 1,000.00	$ 977.70	$ 6.47	$ 1,000.00	$ 1,018.25	$ 6.61	1.32%
Class R4	$ 1,000.00	$ 979.00	$ 5.06	$ 1,000.00	$ 1,019.69	$ 5.16	1.03%
Class R5	$ 1,000.00	$ 980.50	$ 3.53	$ 1,000.00	$ 1,021.22	$ 3.61	0.72%
Class R6	$ 1,000.00	$ 980.70	$ 3.09	$ 1,000.00	$ 1,021.67	$ 3.16	0.63%
Class Y	$ 1,000.00	$ 980.80	$ 3.29	$ 1,000.00	$ 1,021.47	$ 3.36	0.67%
Class F	$ 1,000.00	$ 980.60	$ 3.04	$ 1,000.00	$ 1,021.72	$ 3.11	0.62%
The Hartford Equity Income Fund							
Class A	$ 1,000.00	$ 1,001.90	$ 4.81	$ 1,000.00	$ 1,019.98	$ 4.86	0.97%
Class C	$ 1,000.00	$ 998.50	$ 8.67	$ 1,000.00	$ 1,016.12	$ 8.75	1.75%
Class I	$ 1,000.00	$ 1,003.50	$ 3.63	$ 1,000.00	$ 1,021.17	$ 3.66	0.73%
Class R3	$ 1,000.00	$ 1,000.30	$ 6.74	$ 1,000.00	$ 1,018.05	$ 6.81	1.36%
Class R4	$ 1,000.00	$ 1,001.90	$ 5.21	$ 1,000.00	$ 1,019.59	$ 5.26	1.05%
Class R5	$ 1,000.00	$ 1,003.40	$ 3.78	$ 1,000.00	$ 1,021.03	$ 3.81	0.76%
Class R6	$ 1,000.00	$ 1,003.90	$ 3.23	$ 1,000.00	$ 1,021.57	$ 3.26	0.65%
Class Y	$ 1,000.00	$ 1,003.50	$ 3.63	$ 1,000.00	$ 1,021.17	$ 3.66	0.73%
Class F	$ 1,000.00	$ 1,004.00	$ 3.18	$ 1,000.00	$ 1,021.62	$ 3.21	0.64%
The Hartford Growth Opportunities Fund							
Class A	$ 1,000.00	$ 700.90	$ 4.51	$ 1,000.00	$ 1,019.49	$ 5.36	1.07%
Class C	$ 1,000.00	$ 698.10	$ 7.83	$ 1,000.00	$ 1,015.57	$ 9.30	1.86%
Class I	$ 1,000.00	$ 701.50	$ 3.59	$ 1,000.00	$ 1,020.58	$ 4.26	0.85%
Class R3	$ 1,000.00	$ 699.50	$ 6.11	$ 1,000.00	$ 1,017.60	$ 7.25	1.45%
Class R4	$ 1,000.00	$ 700.40	$ 4.85	$ 1,000.00	$ 1,019.09	$ 5.76	1.15%
Class R5	$ 1,000.00	$ 701.60	$ 3.59	$ 1,000.00	$ 1,020.58	$ 4.26	0.85%
Class R6	$ 1,000.00	$ 701.90	$ 3.12	$ 1,000.00	$ 1,021.12	$ 3.71	0.74%
Class Y	$ 1,000.00	$ 701.60	$ 3.54	$ 1,000.00	$ 1,020.63	$ 4.21	0.84%
Class F	$ 1,000.00	$ 702.00	$ 3.12	$ 1,000.00	$ 1,021.12	$ 3.71	0.74%
The Hartford Healthcare Fund							
Class A	$ 1,000.00	$ 861.40	$ 5.77	$ 1,000.00	$ 1,018.60	$ 6.26	1.25%
Class C	$ 1,000.00	$ 858.10	$ 9.35	$ 1,000.00	$ 1,014.73	$ 10.14	2.03%
Class I	$ 1,000.00	$ 862.40	$ 4.66	$ 1,000.00	$ 1,019.79	$ 5.06	1.01%
Class R3	$ 1,000.00	$ 859.90	$ 7.42	$ 1,000.00	$ 1,016.81	$ 8.05	1.61%
Class R4	$ 1,000.00	$ 861.20	$ 6.05	$ 1,000.00	$ 1,018.30	$ 6.56	1.31%
Class R5	$ 1,000.00	$ 862.40	$ 4.66	$ 1,000.00	$ 1,019.79	$ 5.06	1.01%
Class R6	$ 1,000.00	$ 862.80	$ 4.11	$ 1,000.00	$ 1,020.38	$ 4.46	0.89%
Class Y	$ 1,000.00	$ 862.40	$ 4.62	$ 1,000.00	$ 1,019.84	$ 5.01	1.00%
Class F	$ 1,000.00	$ 863.00	$ 4.11	$ 1,000.00	$ 1,020.38	$ 4.46	0.89%
The Hartford MidCap Fund							
Class A	$ 1,000.00	$ 811.90	$ 4.90	$ 1,000.00	$ 1,019.39	$ 5.46	1.09%
Class C	$ 1,000.00	$ 809.10	$ 8.35	$ 1,000.00	$ 1,015.57	$ 9.30	1.86%
Class I	$ 1,000.00	$ 812.80	$ 3.86	$ 1,000.00	$ 1,020.53	$ 4.31	0.86%
Class R3	$ 1,000.00	$ 810.40	$ 6.51	$ 1,000.00	$ 1,017.60	$ 7.25	1.45%
Class R4	$ 1,000.00	$ 811.70	$ 5.12	$ 1,000.00	$ 1,019.14	$ 5.71	1.14%
Class R5	$ 1,000.00	$ 813.00	$ 3.87	$ 1,000.00	$ 1,020.53	$ 4.31	0.86%
Class R6	$ 1,000.00	$ 813.30	$ 3.33	$ 1,000.00	$ 1,021.12	$ 3.71	0.74%
Class Y	$ 1,000.00	$ 813.00	$ 3.51	$ 1,000.00	$ 1,020.93	$ 3.91	0.78%
Class F	$ 1,000.00	$ 813.20	$ 3.33	$ 1,000.00	$ 1,021.12	$ 3.71	0.74%

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Low choice count: 1 — Haverford Quality Growth Stock (HAVGX)

THE ADVISORS' INNER CIRCLE FUND

HAVERFORD QUALITY GROWTH STOCK FUND (UNAUDITED)

DISCLOSURE OF FUND EXPENSES

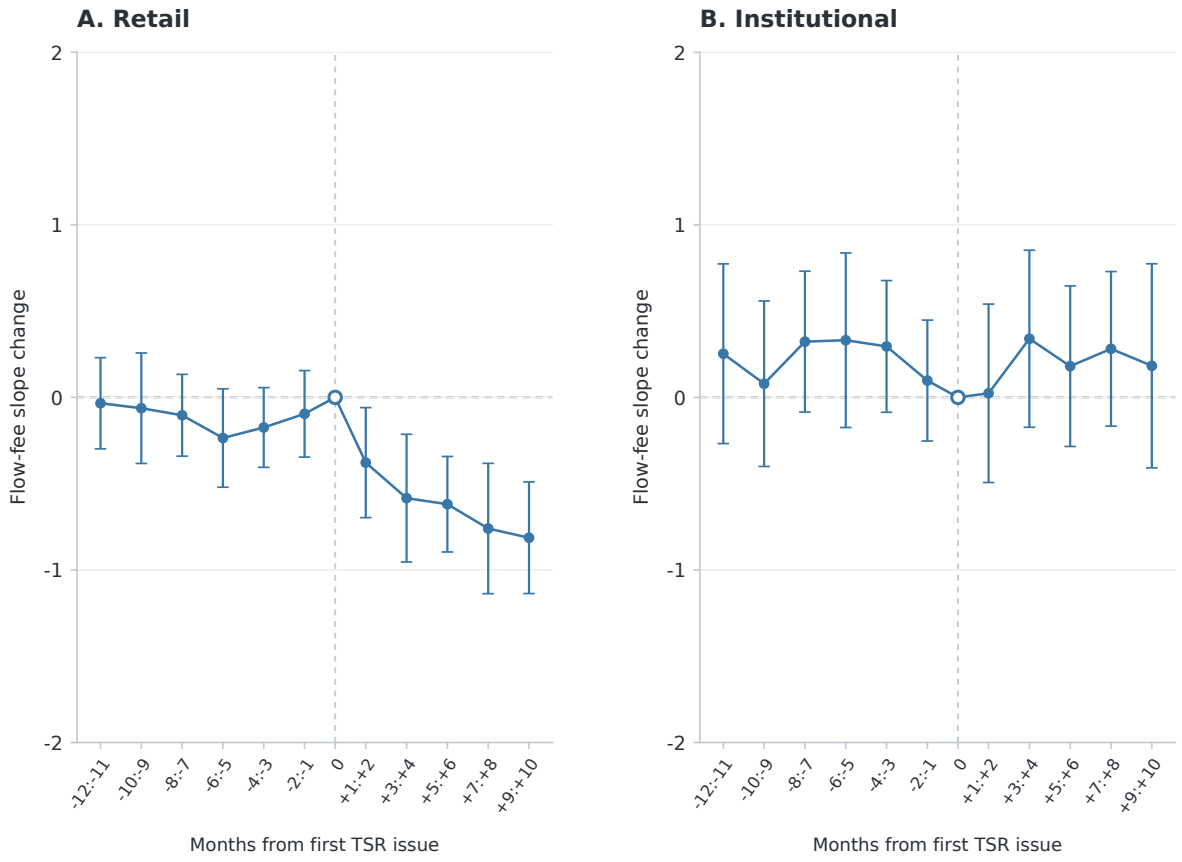
	Beginning Account Value 11/01/21	Ending Account Value 04/30/22	Annualized Expense Ratios	Expenses Paid During Period*
Actual Fund Return	$ 1,000.00	$928.60	0.80%	$3.83
Hypothetical 5% Return	1,000.00	1,020.83	0.80	4.01

* Expenses are equal to the Fund's annualized expense ratio multiplied by the average account value over the period, multiplied by 181/365 (to reflect the one-half year period).

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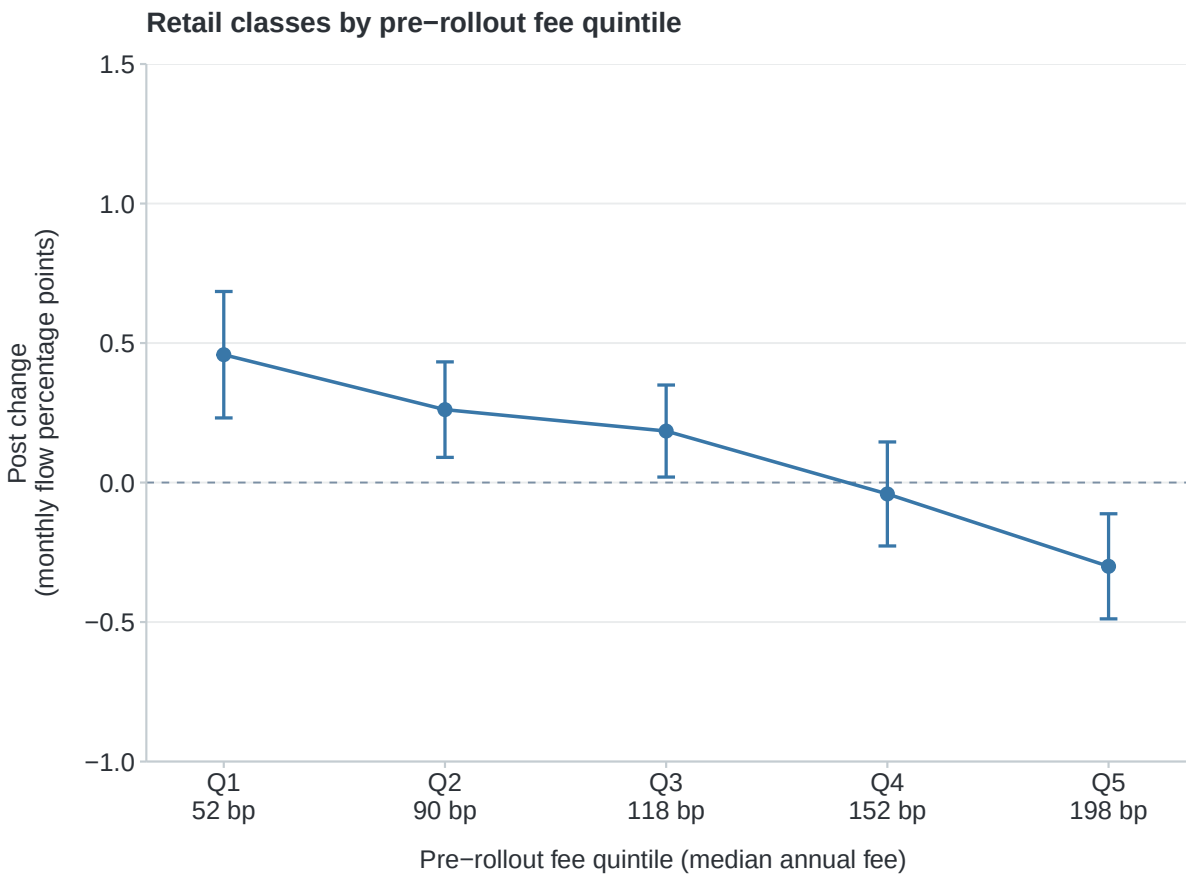
Notes: This figure illustrates variation in the accessibility of expense information in pre-TSR reports. Panel B shows the expense pages from two reports. Hartford's page (page 16) contains five funds with nine classes each, giving 45 distinct fund/class choices. Haverford's page (page 18) contains one fund choice. Actual and hypothetical return scenarios for the same fund or class count once. Report period: April 30, 2022.

Figure 4: Dynamics of flow–fee sensitivity for retail and institutional classes



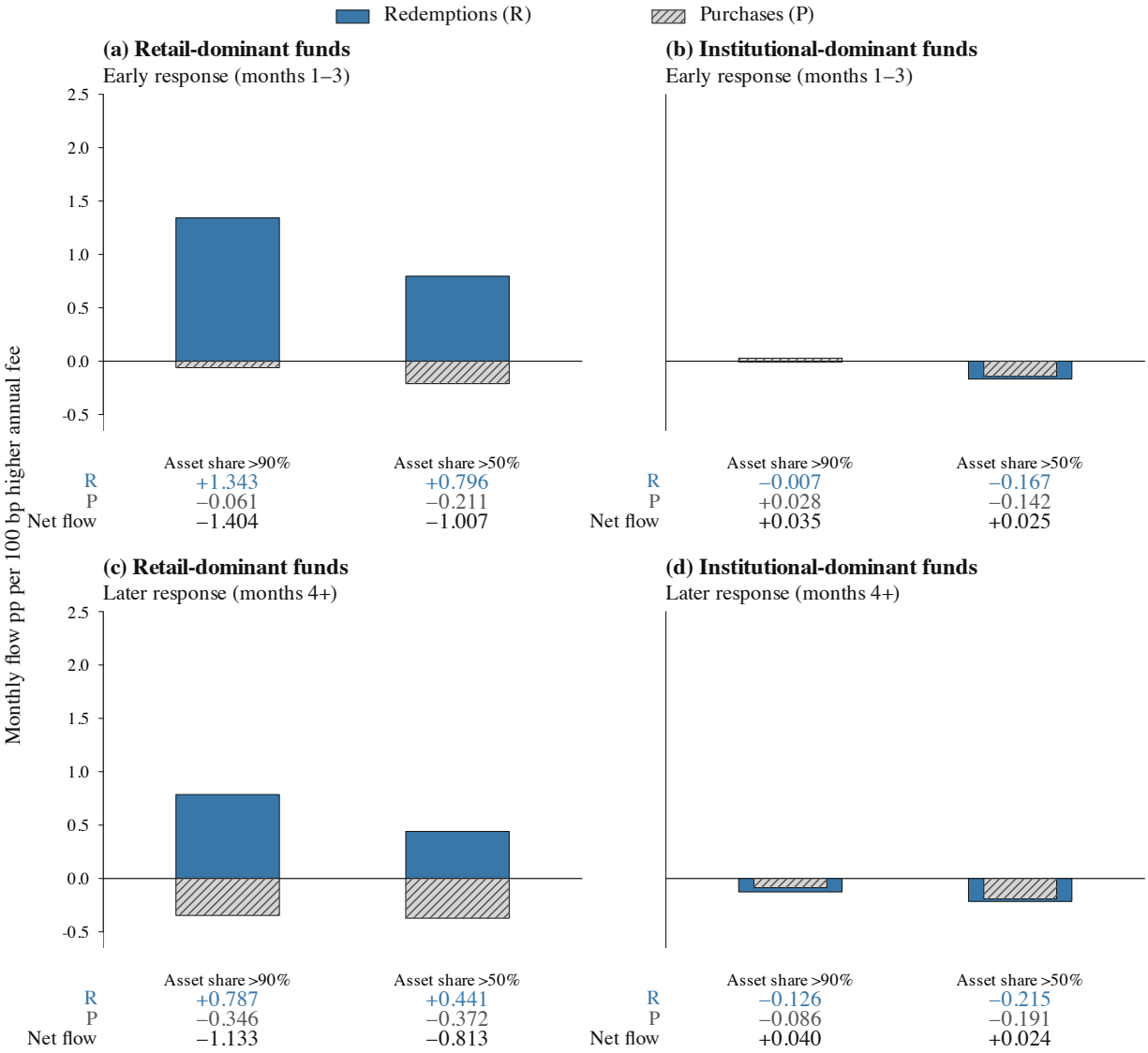
Notes: This figure shows the dynamics of the monthly net-flow–fee relation around the first tailored shareholder report for retail classes (Panel A) and institutional classes (Panel B). Points report changes in the flow–fee slope relative to the issue month (month zero). The other points group event months in two-month intervals. The display covers months −12 through +10. Coefficients are monthly-flow percentage points per 100 basis points of annual expenses. The sample covers July 2023 through July 2025. Both specifications include share-class fixed effects, category-by-month fixed effects, and the controls and Post interactions listed in Table 3, and adjust for a fee trend. 95% confidence intervals are reported based on standard errors clustered by fund family.

Figure 5: Retail net-flow changes across pre-rule fee quintiles



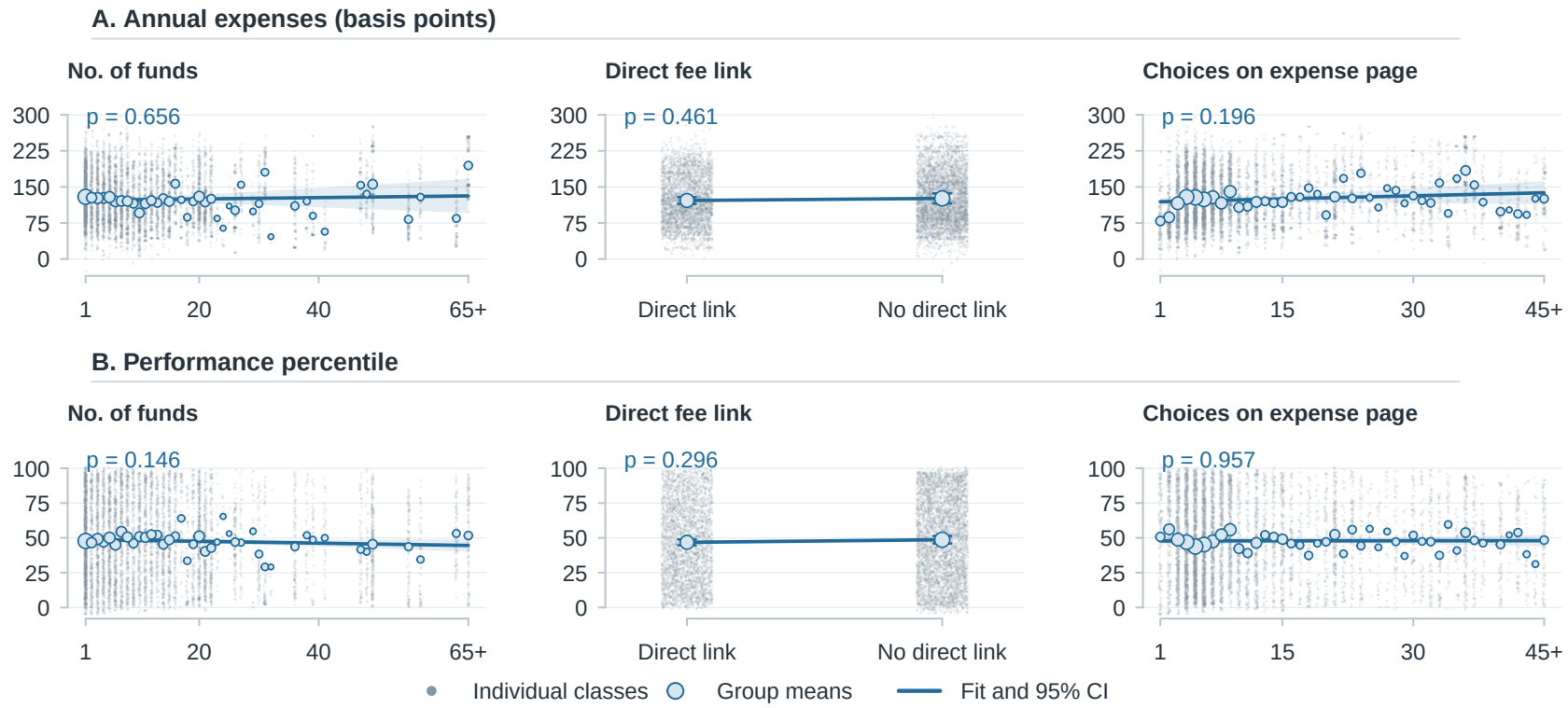
Notes: This figure shows changes in monthly net flow after first TSR issuance for retail classes in each pre-rule fee quintile. Net flow is measured as a percentage of preceding-month assets. Post begins in the month after the issue month. Q1 and Q5 are the lowest- and highest-fee quintiles. The changes are jointly estimated and evaluated at annual net-return rank 0.5 and the model's mean controls. Quintiles use mean annual expenses over October 2021–September 2022 for eligible retail classes with at least six valid fee months. Median expenses in Q1–Q5 are 52, 90, 118, 152, and 198 basis points. The estimation sample covers July 2023–July 2025. The model includes share-class and category-by-month fixed effects and the controls and Post interactions in Table 3, and adjusts for a fee trend. 95% confidence intervals are reported based on standard errors clustered by fund family.

Figure 6: Purchase and redemption responses by fund clientele



Notes: This figure shows changes in redemption–fee sensitivity (blue) and purchase–fee sensitivity (gray hatching). Panels (a)–(b) cover event months +1–+3; panels (c)–(d) cover all observed months from +4 onward. Left panels show retail-dominant funds and right panels institutional-dominant funds. Each panel reports two clientele thresholds, above 90% and above 50% of June 2024 assets. Purchases exclude reinvested distributions; redemptions are positive withdrawals. Both are percentages of prior-month fund assets. Coefficients are monthly-flow percentage points per 100 basis points of lagged annual expenses. Bars retain their signs and overlap when they have the same sign. The displayed "Net flow" is the purchase coefficient minus the redemption coefficient. The sample covers July 2023–July 2025. Models include fund and group-specific category-by-month fixed effects, fund-level versions of the controls in Table 3 and their interactions, and separate purchase and redemption fee trends. Only point estimates are plotted. Table IA.2 reports the full table and standard errors clustered by fund family.

Figure 7: Pre-TSR expense accessibility and fund characteristics



Notes: This figure shows the relation between pre-TSR expense accessibility and fund characteristics. The y-axis shows annual expenses in basis points in Panel A and the prior 12-month category-relative net-return percentile in Panel B. Higher percentiles indicate better performance. From left to right, the x-axes show the number of funds in the filing, whether a verified applicable fee hyperlink is present, and the number of fund/class choices on the expense page. Each subplot estimates a class-level OLS regression of the outcome on the accessibility measure. Faint dots show individual outcomes. Horizontal jitter separates overlapping observations. Blue circles show means at each observed capped count or link status, with area increasing with class count. Shading and whiskers show pointwise 95% confidence intervals clustered by fund family. The p-values test the accessibility slope or link-group difference.

Table 1: Summary statistics for retail and institutional classes

Variable	Mean	SD	P25	Median	P75
Panel A. Retail classes					
Monthly net flow (%)	-0.778	5.109	-2.049	-0.826	0.176
Negative net flow indicator	0.720	0.449	0.000	1.000	1.000
Annual expense ratio (bp)	120.020	56.677	80.000	115.000	160.000
Monthly return (%)	0.990	3.730	-1.333	0.856	3.181
Return volatility (%)	3.674	1.960	2.225	3.511	4.971
Annual net-return rank	0.468	0.289	0.214	0.450	0.713
Log class assets	3.494	2.744	1.629	3.567	5.438
Log class age	5.228	0.690	4.860	5.389	5.727
Log family assets	11.498	2.672	10.125	11.791	13.213
Log family age	6.475	0.592	6.122	6.522	7.003
Post	0.347	0.476	0.000	0.000	1.000
Observations					201,825
Panel B. Institutional classes					
Monthly net flow (%)	-0.292	5.473	-1.710	-0.420	0.830
Negative net flow indicator	0.604	0.489	0.000	1.000	1.000
Annual expense ratio (bp)	69.241	38.498	43.000	69.000	94.000
Monthly return (%)	1.062	3.561	-1.272	0.991	3.251
Return volatility (%)	3.555	1.739	2.268	3.456	4.785
Annual net-return rank	0.536	0.279	0.300	0.548	0.778
Log class assets	3.994	2.853	2.303	4.317	6.019
Log class age	4.850	0.629	4.466	4.890	5.323
Log family assets	12.010	2.253	10.721	12.163	13.321
Log family age	6.485	0.604	6.142	6.491	7.003
Post	0.342	0.474	0.000	0.000	1.000
Observations					256,436

Notes: This table reports share-class-month summary statistics for the separate retail and institutional regression samples in Table 3 from July 2023 through July 2025. Panel A includes retail classes and Panel B institutional classes. Continuous variables are winsorized at the 1st and 99th percentiles separately within each sample, as in the corresponding regressions.

Table 2: Shareholder-report length and pre-TSR expense accessibility

Characteristic	Mean	SD	P25	Median	P75
Words in pre-TSR report	64,857.5	76,002.8	19,791.0	38,962.0	77,375.0
Words per report in first TSR filing	841.6	456.4	503.2	775.3	1,024.3
Funds in pre-TSR filing	14.94	15.72	4	10	20
No verified direct fee link	0.57	0.50	0	1	1
Fund or class choices on expense page	12.95	12.01	5	8	17

Notes: This table describes the length of shareholder reports before and after TSR, and pre-TSR fee accessibility for the retail share classes.

Table 3: Flow–fee sensitivity around the first tailored shareholder report

	All classes		Retail and institutional		Institutional	Retail
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Fee	−0.2908***	−0.2156**	−0.0666	0.0066	−0.2419	−0.5061***
	(0.0900)	(0.0884)	(0.1420)	(0.1437)	(0.1578)	(0.1001)
Post × Fee × Retail			−0.7175***	−0.7156***		
			(0.2030)	(0.2028)		
Post	0.4781	0.4438	0.5512	0.5308	0.5647	0.5691
	(0.5335)	(0.5329)	(0.7092)	(0.7088)	(0.7202)	(0.7392)
Fee	−1.1425***	−1.1749***	−0.8250	−0.8572	−0.9161	−1.2942***
	(0.4172)	(0.4174)	(0.5514)	(0.5513)	(0.6372)	(0.4366)
Annual net-return rank	0.6044***	0.6046***	0.6536***	0.6530***	0.6542***	0.5677***
	(0.0878)	(0.0879)	(0.1209)	(0.1208)	(0.1218)	(0.1135)
Post × Annual net-return rank	1.3243***	1.3229***	1.3781***	1.3775***	1.3954***	1.3844***
	(0.1442)	(0.1439)	(0.1801)	(0.1800)	(0.1829)	(0.1672)
Controls and Post interactions	Yes	Yes	Yes	Yes	Yes	Yes
Fee trend control	No	Yes	No	Yes	Yes	Yes
Share-class FE	Yes	Yes	Yes	Yes	Yes	Yes
Category × month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	466,063	466,063	458,261	458,261	256,436	201,825
Adjusted R^2	0.1373	0.1373	0.1361	0.1362	0.1225	0.1517

Notes: This table reports regressions of monthly share-class net flow, measured as asset growth net of investment returns in percent of preceding-month assets. Fee is the lagged annual expense ratio in percentage points. Post equals one from the month after the first TSR issue month. The sample covers July 2023–July 2025. Columns (1)–(2) include all eligible classes. Columns (3)–(4) jointly estimate retail and institutional classes. Columns (5)–(6) estimate institutional and retail classes separately. All models include share-class and category-by-month fixed effects (FE). Category-by-month FE and control slopes differ by clientele in columns (3)–(4). Controls are lagged monthly return, return volatility, log class assets and age, log family assets and age, and the preceding 12-month net-return rank within category. Each enters separately and in an interaction with Post. Appendix A defines their construction and timing. Table IA.1 reports the control coefficients. Adjusted R^2 measures overall fit including fixed effects. Parentheses contain standard errors clustered by fund family. Significance levels: *0.10, **0.05, ***0.01.

Table 4: Retail flow–fee sensitivity: alternative empirical comparisons

Panel A. Within-fund and category comparisons			
	(1) Portfolio × month	(2) Fund × month	(3) Category-adjusted flow
Post × Fee	−0.9168***	−0.8967***	−0.6294***
	(0.1885)	(0.1889)	(0.1062)
Controls and Post interactions	Yes	Yes	Yes
Fee trend control	Yes	Yes	Yes
Share-class FE	Yes	Yes	Yes
Time effects	Portfolio × month	Fund × month	Month
Observations	155,297	155,238	201,825
Adjusted R^2	0.1837	0.1834	0.1270
Panel B. Alternative time comparisons			
	(4) Fee × calendar quarter	(5) Actual minus historical	(6) Stacked DiD
Post × Fee	−0.3043**	−0.6028***	−0.3358*
	(0.1528)	(0.1430)	(0.1912)
Controls and Post interactions	Yes	Yes	Yes
Fee trend control	No	Yes	Yes
Share-class FE	Yes	Class × episode	Class × stack
Time effects	Category × month	Category × month × episode	Category × month × stack
Observations	201,825	570,932	139,622
Adjusted R^2	0.1520	0.1549	0.2073

Notes: This table reports changes in retail net-flow–fee sensitivity under alternative comparisons. Net flow is measured as a percentage of preceding-month assets. Fee is the lagged annual expense ratio in percentage points. Coefficients are monthly-flow percentage points per 100 basis points of annual expenses. Post begins in the month after the first TSR issue month. Columns (1)–(4) cover July 2023–July 2025. Columns (1) and (2) compare share classes within the same CRSP portfolio or SEC fund series in the same month. Column (3) subtracts the lagged-asset-weighted net flow of other classes in the same category. Column (4) replaces the pre-estimated fee trend with fee-by-calendar-quarter interactions. Column (5) reports the actual Post–fee change minus the average around reporting months shifted back by 24 and 36 months. Column (6) stacks the July and August 2024 issue cohorts against classes issuing after each stack's post window. All models use the controls and Post interactions in Table 3. FE denotes fixed effects. Parentheses contain standard errors clustered by fund family. Significance levels: *0.10, **0.05, ***0.01.

Table 5: Pre-TSR expense accessibility and retail flow–fee sensitivity

Accessibility measure	(1) High filing breadth	(2) No verified fee link	(3) Many expense-page choices
Post × Fee	−0.308***	−0.133	−0.290***
	(0.104)	(0.140)	(0.103)
Post × Fee × Low accessibility	−0.476**	−0.578***	−0.705***
	(0.225)	(0.214)	(0.251)
Controls and Post interactions	Yes	Yes	Yes
Fee trend control	Yes	Yes	Yes
Share-class FE	Yes	Yes	Yes
Category × month FE	Yes	Yes	Yes
Observations	175,011	175,011	175,011
Adjusted R^2	0.1429	0.1388	0.1404

Notes: This table reports how the change in retail flow–fee sensitivity varies with the accessibility of the pre-adoption shareholder report. Net flow is asset growth net of investment returns as a percentage of preceding-month assets. Fee is the lagged annual expense ratio in percentage points, so one unit equals 100 basis points. Post begins in the month after the first TSR issue month. The sample contains retail share-class-months from July 2023 through July 2025. Columns (1) and (3) compare the upper tercile-based group of filing breadth and expense-page choices, respectively, with all other classes. Column (2) compares reports without a verified applicable direct HTML fee link with those having a link. The triple interaction measures the additional change in flow–fee sensitivity for the low-accessibility group. All regressions include share-class and category-by-month fixed effects, the controls and Post interactions in Table 3, and a fee trend. Parentheses contain standard errors clustered by fund family. Significance levels: *0.10, **0.05, ***0.01.

Table 6: Recent performance changes and retail flow–fee sensitivity

	Entered bottom tercile		Entered top tercile	
Dependent variable: monthly net flow (%)	(1)	(2)	(3)	(4)
Post × Fee	−0.670***	−0.433***	−0.789***	−0.553***
	(0.154)	(0.114)	(0.149)	(0.111)
Post × Fee × Entry	−0.356**	−0.354**	0.318**	0.323**
	(0.158)	(0.158)	(0.146)	(0.146)
Controls and Post interactions	Yes	Yes	Yes	Yes
Fee trend control	No	Yes	No	Yes
Share-class FE	Yes	Yes	Yes	Yes
Category-by-month FE	Yes	Yes	Yes	Yes
Observations	201,047	201,047	201,047	201,047
Adjusted R^2	0.1516	0.1518	0.1516	0.1519

Notes: This table reports changes in retail flow–fee sensitivity for classes entering the bottom or top tercile of relative performance. Net flow is a percentage of preceding-month assets, and Fee is the lagged annual expense ratio in percentage points. Post begins in the month after the first TSR issue month. The sample consists of retail share-class-months from July 2023 through July 2025. Annual net returns are ranked within Lipper categories over July 2022–June 2023 and July 2023–June 2024. Columns (1)–(2) identify classes that enter the bottom tercile in the recent year after being outside it in the earlier year. Columns (3)–(4) identify classes that enter the top tercile after being outside it in the earlier year. Each group is compared with all other retail classes. Entry is the indicator for entering the tercile named in the column header. Post × Fee is the slope change for the comparison group, and its interaction with Entry is the additional change for the entering group. All regressions retain the baseline lagged monthly return and annual net-return rank controls and their Post interactions. They also include share-class and category-by-month fixed effects, the remaining controls and Post interactions in Table 3, and the lower-order terms involving Entry. Columns (2) and (4) also include the retail fee trend estimated over July 2019–June 2024. Parentheses contain standard errors clustered by fund family. Significance levels: *0.10, **0.05, ***0.01.

Table 7: Changes in retail expenses by fee component

Dependent variable: annual fee (bp)	(1) Total expenses	(2) 12b-1 fees	(3) Nonmarketing
Post-announcement × Q5	−3.48***	0.75	−4.19***
	(1.09)	(0.54)	(1.21)
Controls and Post interactions	Yes	Yes	Yes
Fee trend control	Yes	Yes	Yes
Share-class FE	Yes	Yes	Yes
Style × year FE	Yes	Yes	Yes
Observations	30,706	30,706	30,706
Adjusted R^2	0.9851	0.9943	0.9733

Notes: This table reports changes in retail share-class expenses around the October 26, 2022 adoption announcement. The dependent variables are annual total expenses, actual 12b-1 marketing and distribution fees, and nonmarketing expenses (total expenses minus actual 12b-1 fees), in basis points (bp). The sample covers fiscal-period endpoints in 2020–2025 and requires observed fee components and controls. Q5 and Q1 are the highest and lowest expense quintiles, formed separately for active and passive funds within each investment style using mean October 2021–September 2022 expense ratios in the retail source population and held fixed thereafter. Post-announcement pools the first two full fiscal years after October 26, 2022. Earlier fiscal years, including the year containing the announcement, form the pre-period reference. All quintiles remain in estimation, with Q1 as the reference group. The reported coefficient measures the post-announcement change in the Q5–Q1 gap. Relative fee-quintile linear trends are estimated over October 2017–September 2022 and subtracted before estimation. All regressions include share-class and style-by-year fixed effects. Controls are annual counterparts to those in Table 3, measured at the last completed month before each annual fee period begins. Controls enter separately and in interactions with Post-announcement. Parentheses contain standard errors clustered by fund family. Significance levels: *0.10, **0.05, ***0.01.

Table 8: Portfolio differentiation by clientele

	Retail-dominant		Institutional-dominant	
	(1) Own-benchmark Active Share (pp)	(2) Own-index tracking error (annualized pp)	(3) Own-benchmark Active Share (pp)	(4) Own-index tracking error (annualized pp)
Transition × Q5	0.305	0.564***	−0.059	0.162
	(0.340)	(0.201)	(0.495)	(0.222)
Post-implementation × Q5	1.446**	0.750***	0.288	0.526**
	(0.556)	(0.280)	(0.720)	(0.221)
Fund controls	Yes	Yes	Yes	Yes
Portfolio FE	Yes	Yes	Yes	Yes
Style × time FE	Yes	Yes	Yes	Yes
Time unit	Quarter	Month	Quarter	Month
Observations	2,617	10,004	4,798	21,434
Adjusted R^2	0.9767	0.7799	0.9785	0.7960

Notes: This table reports changes in portfolio differentiation by pre-rule expenses and clientele. The dependent variables are Active Share in percentage points and tracking error in annualized percentage points. Active Share is one-half the sum of absolute differences between company-level portfolio and disclosed-benchmark weights, using normalized common-stock holdings. Tracking error is the annualized standard deviation of net fund returns minus dividend-inclusive own-index returns over 12 complete trailing months. The sample includes active domestic equity portfolios. An investment portfolio can be shared by multiple funds and their share classes, including both retail and institutional classes. Retail-dominant and institutional-dominant portfolios have more than 50% of September 2022 assets in the corresponding classes. At least 80% of portfolio assets must have known class designations. Each portfolio's pre-rule expense ratio is an average across all its share classes, weighted by September 2022 assets. Within each investment style, retail-dominant and institutional-dominant portfolios are ranked separately by this expense ratio. Q5 is the highest-fee quintile; the other four quintiles form the comparison group. Transition covers 2022Q4–2024Q2, and Post-implementation covers 2024Q3–2025Q3. The reported coefficients measure changes in the Q5-minus-pooled-Q1–Q4 outcome gap relative to the pre-announcement period, 2021Q3–2022Q3. Controls are log fund size (portfolio assets), log fund age, and log family size (management-family assets). Appendix A gives construction details. Parentheses contain standard errors clustered by fund family. Significance levels: *0.10, **0.05, ***0.01.

Appendix A Variable Definitions

The table defines the principal variables used in the main analysis. Unless otherwise indicated, monthly share-class data are from CRSP. Variables specific to Internet Appendix analyses are defined in the corresponding exhibit notes.

Variable	Definition
Flow outcomes	
Net flow	Monthly change in class assets net of investment returns, divided by assets in the preceding observed month and multiplied by 100; see Equation (4).
Purchases	Monthly Form N-PORT purchases, excluding reinvestment, divided by prior-calendar-month whole-fund assets and multiplied by 100.
Redemptions	Monthly Form N-PORT redemptions, recorded as positive outflows, divided by prior-calendar-month whole-fund assets and multiplied by 100.
Flow regressions: fee, timing, and clientele	
Fee	Annual expense ratio for the preceding observed class-month, in percentage points. One unit equals 100 basis points of annual expenses.
Post	Indicator equal to one beginning in the first month after the issue month. Event month is the calendar-month difference from the issue month; month zero remains in the pre-period in the static flow regressions.
Issue month	Month assigned to the first successfully extracted and linked TSR filing. Filings on days 1–14 are assigned to the preceding month and later filings to the filing month.
Fund issue month	Issue month assigned using the earliest linked TSR filing among valid mapped classes.
Retail and institutional	Designations frozen at the latest actual CRSP observation in October 2021–September 2022. Retail requires retail "Y" and institutional "N"; institutional requires retail "N" and institutional "Y." Other or missing combinations are unknown.
Monthly share-class controls	
Monthly return	Monthly net return.
Log class assets	Natural logarithm of class assets in millions of dollars.
Return volatility	Standard deviation of net returns over 12 observed class-months.
Log class age	Natural logarithm of completed months since the class's first-offer date.

Variable definitions (continued)

Variable	Definition
Log family assets	Natural logarithm of total assets in millions of dollars across classes in the same management-code family, constructed before report-linkage and regression-sample restrictions.
Log family age	Natural logarithm of completed months since the earliest first-offer date among family members observed by that date.
Annual net-return rank	Rank of the class's compounded net return over the preceding 12 calendar months within its dated Lipper category, divided by the number of valid classes. Twelve consecutive valid returns and at least 20 classes are required. Ranks use the full dated non-money-market CRSP population, with average ranks for ties.
Pre-TSR report accessibility and performance history	
Filing breadth	Number of distinct SEC series in the filing.
No direct fee link	Indicator for a report without a verified applicable HTML hyperlink to the class's expense information.
Expense-page choices	Number of distinct fund or class entries across expense tables on the publisher's page containing the applicable expense.
Entry into bottom/top tercile	Indicators for entering the bottom ($\leq 1/3$) or top ($\geq 2/3$) tercile of category-relative annual net returns in July 2023–June 2024 after being outside that tercile in July 2022–June 2023. Both ranks are fixed in June 2024 and must be observed.
Fund responses	
Expense components	Annual total expenses and actual 12b-1 fees, in basis points. Nonmarketing expenses equal total expenses minus actual 12b-1 fees before winsorization.
Active Share	One-half the sum of absolute differences between portfolio and disclosed-benchmark weights, aggregated to companies and normalized within each common-stock sleeve, multiplied by 100 and expressed in percentage points. Benchmark identities are held fixed under continuity; constituent weights are dated.
Tracking error	$100\sqrt{12}$ times the standard deviation of net fund returns minus dividend-inclusive own-index returns over 12 complete trailing calendar months, including the current month, in annualized percentage points.
Portfolio fee quintiles	Fixed quintiles of pre-rule all-class portfolio expenses, ranked within clientele, frozen investment style, and passive status on the full reference population before outcome restrictions. Expenses use October 2021–September 2022 class averages and September 2022 asset weights.

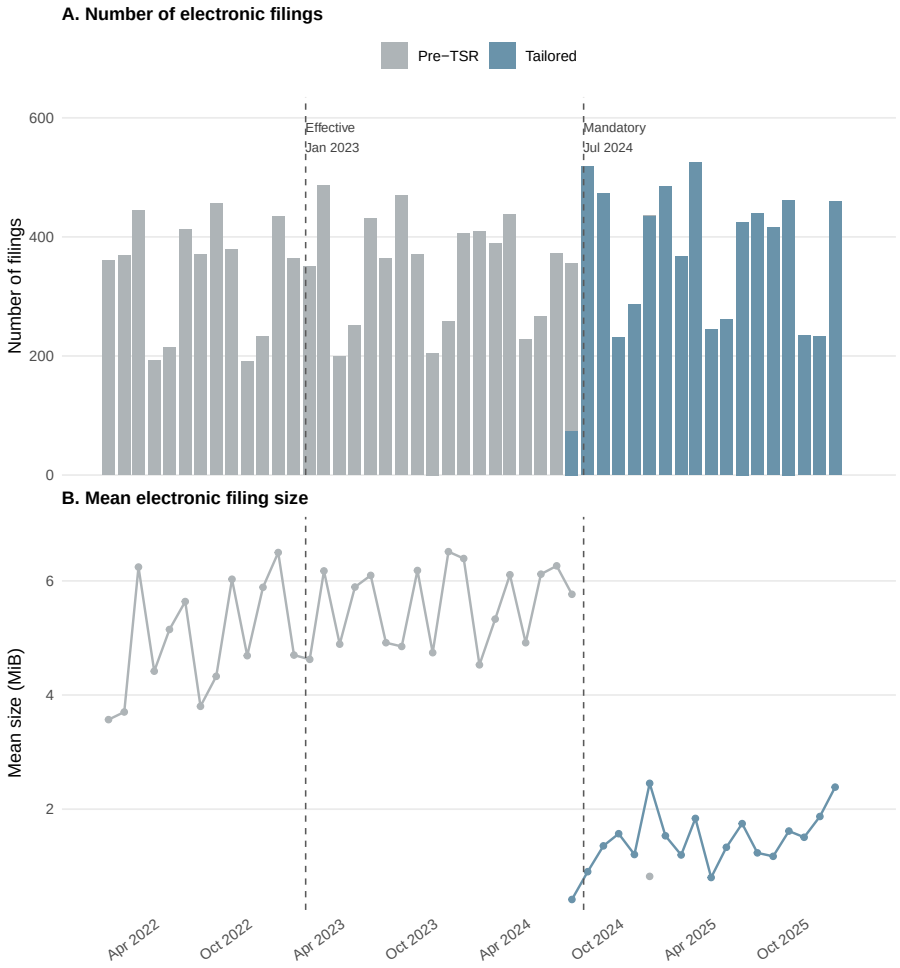
Variable definitions (continued)

Variable	Definition
Portfolio-response phases	Pre-announcement reference: 2021Q3–2022Q3; post-announcement stages: transition, 2022Q4–2024Q2, and post-implementation, 2024Q3–2025Q3. Changes in the Q5-minus-pooled-Q1–Q4 gap in each stage are measured relative to the reference period, separately by clientele and outcome.
Log portfolio assets	Natural logarithm of observed total assets in millions of dollars across all dated classes mapped to the investment portfolio. Table 8 uses the previous quarter-end value for Active Share and the previous month-end value for tracking error. Aggregation precedes outcome restrictions.
Log portfolio age	Natural logarithm of positive completed months since the earliest class inception known in the portfolio's dated membership history by the observation date. Measured at the current date in Table 8; CRSP inception fallbacks and the available mapping history apply.
Portfolio family assets	Total assets in millions of dollars across the full dated CRSP management-code family, before report or outcome restrictions. Table 8 includes its natural logarithm at the same lag as portfolio assets and requires an unambiguous family assignment. Asset sums retain observed values and do not impute missing class assets.
Retail and institutional asset shares	September 2022 assets in each strict class designation divided by all portfolio assets, including assets with unknown designations. A portfolio is retail-dominant or institutional-dominant when the corresponding asset share exceeds 50%, with at least 80% known-designation coverage.

Internet Appendix

IA.1 Shareholder Report Filings and File Size

Figure IA.1: Shareholder report filings and electronic file size



Notes: This figure shows the monthly number and size of linked shareholder-report filings. Panel A shows the monthly number of N-CSR(S) filings by report format. Panel B shows the mean size of the extracted shareholder reports in mebibytes (MiB). The sample covers January 2022–December 2025 and includes filings linked to share classes with at least one pre-TSR report and one tailored report in that window. Dashed lines mark the January 2023 effective date and the July 2024 mandatory compliance date. Electronic files include HTML markup and may contain information for multiple funds or classes.

IA.2 Baseline Results with Controls

Table IA.1: Baseline flow–fee regressions with control coefficients

	All classes		Retail and institutional		Institutional	Retail
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Fee	−0.2908***	−0.2156**	−0.0666	0.0066	−0.2419	−0.5061***
	(0.0900)	(0.0884)	(0.1420)	(0.1437)	(0.1578)	(0.1001)
Post × Fee × Retail			−0.7175***	−0.7156***		
			(0.2030)	(0.2028)		
Post	0.4781	0.4438	0.5512	0.5308	0.5647	0.5691
	(0.5335)	(0.5329)	(0.7092)	(0.7088)	(0.7202)	(0.7392)
Fee	−1.1425***	−1.1749***	−0.8250	−0.8572	−0.9161	−1.2942***
	(0.4172)	(0.4174)	(0.5514)	(0.5513)	(0.6372)	(0.4366)
Lagged monthly return	6.1823***	6.1731***	1.9270	1.9248	2.0135	9.2797***
	(1.7036)	(1.7007)	(1.7878)	(1.7880)	(1.8039)	(2.4090)
Log class assets	−3.0011***	−3.0034***	−2.3136***	−2.3135***	−2.3580***	−4.5272***
	(0.2788)	(0.2793)	(0.1622)	(0.1623)	(0.1648)	(0.7345)
Return volatility	16.7515***	16.9609***	24.6927***	24.8714***	24.4471***	8.6442
	(4.5418)	(4.5355)	(5.0286)	(5.0286)	(5.0271)	(7.2270)
Log class age	1.0723**	1.0314**	0.0446	0.0086	0.1585	3.7408***
	(0.5245)	(0.5175)	(0.6922)	(0.6904)	(0.6924)	(1.2014)
Log family assets	0.1973**	0.1973**	0.1177	0.1172	0.1255	0.3619***
	(0.0989)	(0.0984)	(0.1058)	(0.1056)	(0.1088)	(0.1305)
Log family age	0.3227	0.3209	0.7961***	0.7930***	0.7686***	−0.3760
	(0.4216)	(0.4169)	(0.2275)	(0.2256)	(0.2547)	(1.0280)
Annual net-return rank	0.6044***	0.6046***	0.6536***	0.6530***	0.6542***	0.5677***
	(0.0878)	(0.0879)	(0.1209)	(0.1208)	(0.1218)	(0.1135)
Post × Lagged monthly return	2.0678	2.0830	0.9946	1.0037	1.0276	2.4633
	(1.7495)	(1.7489)	(2.8760)	(2.8775)	(2.9420)	(1.8239)
Post × Log class assets	−0.0736***	−0.0739***	−0.0857***	−0.0861***	−0.0849***	−0.0573**
	(0.0168)	(0.0168)	(0.0211)	(0.0211)	(0.0214)	(0.0238)
Post × Return volatility	−4.6922	−4.9932	−6.2850	−6.5665	−5.6164	1.3677
	(3.1896)	(3.1872)	(4.0191)	(4.0303)	(4.1368)	(4.5245)
Post × Log class age	−0.0279	−0.0328	0.0051	0.0015	0.0140	−0.1473
	(0.0610)	(0.0611)	(0.1121)	(0.1119)	(0.1144)	(0.0895)
Post × Log family assets	0.0086	0.0087	0.0361	0.0362	0.0360	−0.0460
	(0.0220)	(0.0220)	(0.0286)	(0.0286)	(0.0292)	(0.0300)
Post × Log family age	−0.0678	−0.0677	−0.1938	−0.1943	−0.1899	0.1389
	(0.1086)	(0.1084)	(0.1336)	(0.1333)	(0.1351)	(0.1190)
Post × Annual net-return rank	1.3243***	1.3229***	1.3781***	1.3775***	1.3954***	1.3844***
	(0.1442)	(0.1439)	(0.1801)	(0.1800)	(0.1829)	(0.1672)

Continued on next page

Table IA.1 continued

	All classes		Retail and institutional		Institutional	Retail
	(1)	(2)	(3)	(4)	(5)	(6)
Controls and Post interactions	Yes	Yes	Yes	Yes	Yes	Yes
Fee trend control	No	Yes	No	Yes	Yes	Yes
Share-class FE	Yes	Yes	Yes	Yes	Yes	Yes
Category × month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	466,063	466,063	458,261	458,261	256,436	201,825
Adjusted R^2	0.1373	0.1373	0.1361	0.1362	0.1225	0.1517

Notes: This table reports the baseline flow–fee coefficients and the coefficients on all seven controls from Table 3 and their Post interactions. The outcome is monthly net flow as a percentage of preceding-month assets. Fee is the lagged annual expense ratio in percentage points; one unit equals 100 basis points. The sample covers July 2023–July 2025. Post begins in the month after the first tailored-report issue month. Columns (1)–(2) include all eligible classes; columns (3)–(4) jointly estimate retail and institutional classes. Columns (5)–(6) estimate institutional and retail classes separately. Parentheses contain standard errors clustered by fund family. Significance levels: *0.10, **0.05, ***0.01.

IA.3 Purchase and Redemption Responses

Table IA.2: Fee sensitivity of redemptions and purchases over time

	Retail-dominant funds			Institutional-dominant funds		
	(1)	(2)	(3)	(4)	(5)	(6)
	Redemptions	Purchases	Implied net flow	Redemptions	Purchases	Implied net flow
Panel A. Designated asset share $>50\%$						
Months +1–+3 × Fee	0.796*	−0.211	−1.007**	−0.167	−0.142	0.025
	(0.456)	(0.465)	(0.431)	(0.205)	(0.331)	(0.353)
Months +4 onward × Fee	0.441	−0.372	−0.813*	−0.215	−0.191	0.024
	(0.360)	(0.540)	(0.456)	(0.208)	(0.309)	(0.340)
Controls and Post interactions	Yes	Yes	Yes	Yes	Yes	Yes
Fee trend control	Yes	Yes	Yes	Yes	Yes	Yes
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Category × month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	36,987	36,987	36,987	63,324	63,324	63,324
Funds	1,785	1,785	1,785	3,126	3,126	3,126
Adjusted R^2	0.8559	0.8026	—	0.2597	0.2985	—
Panel B. Designated asset share $>90\%$						
Months +1–+3 × Fee	1.343**	−0.061	−1.404**	−0.007	0.028	0.035
	(0.591)	(0.645)	(0.618)	(0.295)	(0.447)	(0.506)
Months +4 onward × Fee	0.787*	−0.346	−1.133*	−0.126	−0.086	0.040
	(0.471)	(0.759)	(0.636)	(0.271)	(0.403)	(0.424)
Controls and Post interactions	Yes	Yes	Yes	Yes	Yes	Yes
Fee trend control	Yes	Yes	Yes	Yes	Yes	Yes
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Category × month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	22,587	22,587	22,587	38,092	38,092	38,092
Funds	1,047	1,047	1,047	1,864	1,864	1,864
Adjusted R^2	0.8556	0.8036	—	0.2783	0.3257	—

Notes: This table reports changes in fund-level redemption–fee and purchase–fee slopes over July 2023–July 2025. Redemptions are positive withdrawals, and purchases exclude reinvested distributions. Both are percentages of prior-month fund assets. Fee is the lagged annual expense ratio in percentage points, so coefficients are monthly-flow percentage points per 100 basis points of annual expenses. Months +1–+3 and +4 onward are measured from the first TSR issue month. The reference period includes all observed months through the issue month. Funds have fully known June 2024 class designations and more than 50% (Panel A) or 90% (Panel B) of assets in the indicated clientele. Controls are those in Table 3, aggregated with fees using lagged class-asset weights and requiring valid inputs for at least 95% of mapped fund assets. They enter separately and in interactions with Post. Models include fund and category-by-month fixed effects (FE), with slopes and time effects specific to each clientele group. Purchase and redemption fee trends are estimated separately over July 2019–June 2024. Adjusted R^2 measures overall fit including FE within each displayed group. Implied net flow is the purchase-minus-redemption coefficient contrast, with standard errors calculated from their joint covariance. Parentheses contain standard errors clustered by fund family. Significance levels: *0.10, **0.05, ***0.01.

IA.4 Documented Early Rule 30e-3 Intent

Rule 30e-3 allowed funds to send investors a notice directing them to an online report. The comparison tests whether documented early intent to use this option predicts a different retail response to TSR. Table IA.3 reports the average flow–fee slope change and the additional change for early-notice funds. The additional change is small in both samples. Matching substantially reduces the average slope change, and its residual imbalance and differential pretrends limit a delivery interpretation.

Table IA.3: Retail flow–fee sensitivity by documented early Rule 30e-3 intent

	Unmatched		Matched	
Dependent variable: monthly net flow (%)	(1)	(2)	(3)	(4)
Average Post × Fee	−0.719***	−0.482***	−0.223	0.002
	(0.140)	(0.103)	(0.189)	(0.176)
Post × Fee × Early notice	−0.059	−0.057	0.004	0.006
	(0.221)	(0.223)	(0.272)	(0.272)
Controls and Post interactions	Yes	Yes	Yes	Yes
Fee trend control	No	Yes	No	Yes
Share-class FE	Yes	Yes	Yes	Yes
Category-by-month FE	Yes	Yes	Yes	Yes
Observations	196,344	196,344	32,746	32,746
Fund families	426	426	180	180
Early-notice observation share	0.930	0.930	0.716	0.716

Notes: This table tests whether the change in retail flow–fee sensitivity varies with documented early intent to use Rule 30e-3. Early notice identifies funds with qualifying 2019–2020 intent or advance-notice evidence. This indicator does not verify delivery arrangements immediately before TSR. Columns (1)–(2) use the unmatched sample. Columns (3)–(4) use equal observation weights in a sample matched on pre-October 2022 characteristics within category and index-status strata. All columns include share-class and common category-by-month fixed effects (FE), the controls in Table 3, and their Post interactions. The sample covers July 2023–July 2025. Net flow is a percentage of preceding-month assets; Fee is the lagged annual expense ratio in percentage points. Post begins in the month after the first TSR issue month. Columns (2) and (4) adjust for the fee trend. Standard errors in parentheses are clustered by fund family. Significance levels: *0.10, **0.05, ***0.01.